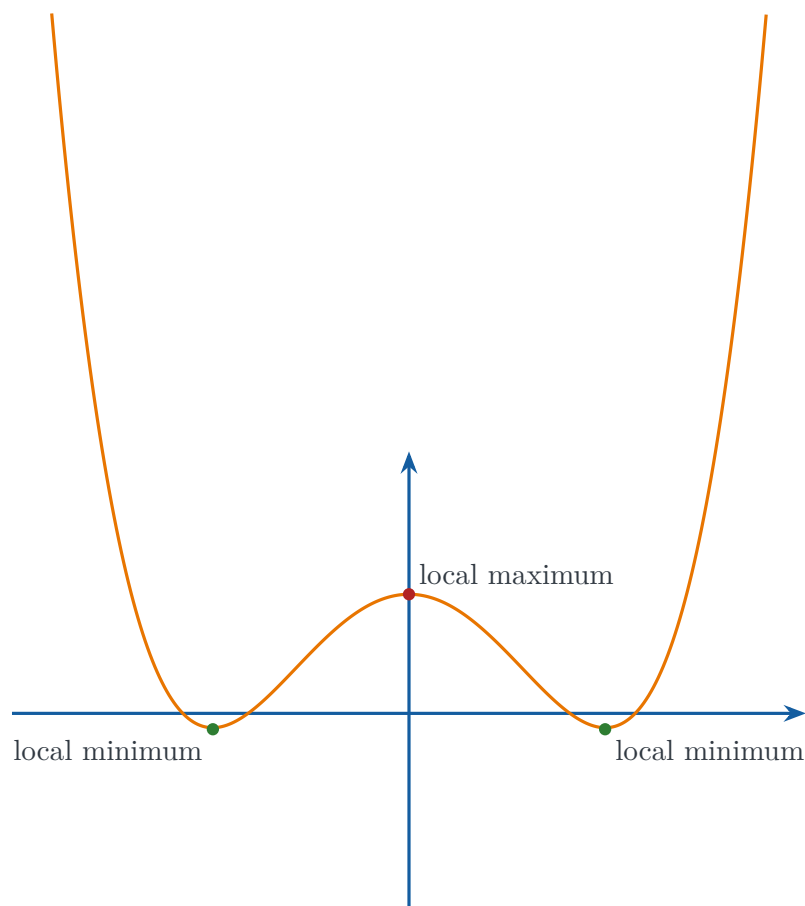


Extrema, Monotonicity, Concavity and Complete Curve Sketching

A Coloured LaTeX Learning Resource

Based on PDF pages **40–78** of the Grade 12 Advanced Mathematics textbook, covering textbook pages **250–288**.

Sections included: **4.3 Maximum and Minimum Values**, **4.4 Increasing and Decreasing Functions**, **4.5 Concavity and the Second Derivative Test**, **4.6 Overview of Curve Sketching**, and the opening of **4.7 Optimization**.



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Grade 12 Advanced Mathematics

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The main body retypesets and expands the definitions, theorems, notes, methods, and all worked examples in the assigned page range. The final appendix reproduces the complete assigned textbook pages, ensuring that every original exercise, application, exploratory task, graph, and diagram is retained without omission.

Learning Objectives

By the end of this unit, students should be able to:

- distinguish absolute and local extrema and identify critical numbers;
- apply the Extreme Value Theorem and the closed-interval method;
- use the sign of f' to identify intervals of increase/decrease and classify extrema;
- use f'' to identify concavity, inflection points, and local extrema;
- combine domain, limits, asymptotes, intercepts, and derivative information to sketch representative graphs;
- recognise when calculator windows hide important local or global behaviour;
- begin translating an applied maximum/minimum problem into a one-variable optimisation model.

Lesson 4.3: Maximum and Minimum Values

The Central Question

Where can a function attain its largest and smallest values, and how can calculus locate them efficiently?

Absolute Extrema

Let f be defined on a set $S \subseteq \mathbb{R}$ and let $c \in S$.

(i) $f(c)$ is an **absolute maximum** on S if $f(c) \geq f(x)$ for every $x \in S$.

(ii) $f(c)$ is an **absolute minimum** on S if $f(c) \leq f(x)$ for every $x \in S$.

Either one is called an **absolute extremum**; together they are **absolute extrema**.

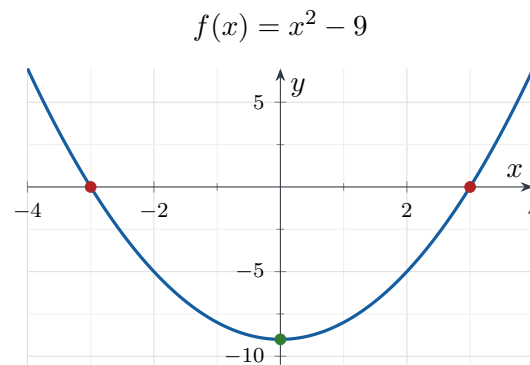
A function need not possess an absolute maximum or minimum. Open endpoints, unbounded domains, and discontinuities can prevent extrema from being attained.

For $f(x) = x^2 - 9$, locate the absolute extrema on:

- $(-\infty, \infty)$,
- $(-3, 3)$,
- $[-3, 3]$.

The parabola opens upward and has vertex $(0, -9)$.

- On $\mathbb{R}$, $f(0) = -9$ is the absolute minimum; there is no absolute maximum.
- On $(-3, 3)$, $f(0) = -9$ remains the absolute minimum. The value 0 is approached near $x = \pm 3$ but is never attained, so there is no absolute maximum.
- On $[-3, 3]$, $f(0) = -9$ is the absolute minimum and $f(-3) = f(3) = 0$ is the absolute maximum.



Locate the absolute extrema of $f(x) = \frac{1}{x}$ on $[-3, 0) \cup (0, 3]$.

As $x \rightarrow 0^+$, $f(x) \rightarrow +\infty$, and as $x \rightarrow 0^-$, $f(x) \rightarrow -\infty$. Therefore the function is unbounded above and below on the stated domain and has neither an absolute maximum nor an absolute minimum.

If f is continuous on a closed, bounded interval $[a, b]$, then f attains both an absolute maximum and an absolute minimum on $[a, b]$.

The theorem gives a *guarantee*, not a necessary condition. A function can have absolute extrema even when it is not continuous or when its domain is not a closed interval.

Find the absolute extrema of $f(x) = \frac{1}{x}$ on $[1, 3]$.

The function is continuous on $[1, 3]$, so the Extreme Value Theorem applies. Since $1/x$ decreases on this interval,

$f(1) = 1$ is the absolute maximum,

$f(3) = \frac{1}{3}$ is the absolute minimum.

Local Extrema and Critical Numbers

A value $f(c)$ is a **local maximum** if $f(c) \geq f(x)$ for all x in some open interval containing c . It is a **local minimum** if $f(c) \leq f(x)$ throughout some open interval containing c .

A number c in the domain of f is a **critical number** if

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

A number at which f' is undefined is a critical number only when that number belongs to the domain of f .

Locate the local extrema of $f(x) = 9 - x^2$ and describe the derivative there.

The vertex is $(0, 9)$, so $f(0) = 9$ is a local (and absolute) maximum. Since $f'(x) = -2x$, we have $f'(0) = 0$ and the tangent line is horizontal.

Locate the local extrema of $f(x) = |x|$.

The graph has a corner at $(0, 0)$, and $f(0) = 0$ is a local and absolute minimum. The derivative is undefined at $x = 0$.

If $f(c)$ is a local extremum and f is defined at c , then c must be a critical number of f .

Fermat's Theorem says that local extrema can occur only at critical numbers. It does *not* say that every critical number produces a local extremum.

Proof idea for Fermat's Theorem

Suppose f is differentiable at c and $f'(c) > 0$. For sufficiently small $h > 0$,

$$\frac{f(c+h) - f(c)}{h} > 0 \implies f(c+h) > f(c),$$

so $f(c)$ cannot be a local maximum. For sufficiently small $h < 0$, the same positive quotient forces $f(c+h) < f(c)$, so $f(c)$ cannot be a local minimum. The case $f'(c) < 0$ is analogous. Therefore a differentiable local extremum must satisfy $f'(c) = 0$.

Find the critical numbers and local extrema of

$$f(x) = 2x^3 - 3x^2 - 12x + 5.$$

$$f'(x) = 6x^2 - 6x - 12 = 6(x - 2)(x + 1).$$

Thus the critical numbers are $x = -1$ and $x = 2$. The graph (or the sign of f') shows a local maximum at

$$(-1, f(-1)) = (-1, 12)$$

and a local minimum at

$$(2, f(2)) = (2, -15).$$

Find the critical numbers and local extrema of

$$f(x) = (3x + 1)^{2/3}.$$

$$f'(x) = \frac{2}{(3x + 1)^{1/3}}.$$

The derivative is never zero, but it is undefined at $x = -\frac{1}{3}$. Since $-\frac{1}{3}$ is in the domain of f , it is a critical number. The graph has a cusp and an absolute minimum at

$$\left(-\frac{1}{3}, 0\right).$$

For $f(x) = x^3$, the only critical number is $x = 0$ because $f'(x) = 3x^2$. Nevertheless, the function increases through $x = 0$, so $(0, 0)$ is not a local extremum.

For $f(x) = x^{1/3}$,

$$f'(x) = \frac{1}{3}x^{-2/3}$$

is undefined at $x = 0$. The graph has a vertical tangent there but continues increasing, so $(0, 0)$ is not a local extremum.

Find the critical numbers of

$$f(x) = \frac{2x^2}{x + 2}.$$

$$f'(x) = \frac{4x(x+2) - 2x^2}{(x+2)^2} = \frac{2x(x+4)}{(x+2)^2}.$$

The derivative is zero at $x = 0, -4$ and undefined at $x = -2$. However, $x = -2$ is not in the domain of f , so the critical numbers are

$$x = -4 \text{ and } x = 0.$$

The Closed-Interval Method

If f is continuous on $[a, b]$, every absolute extremum occurs either at an endpoint or at a critical number in (a, b) .

1. Find every critical number of f in $[a, b]$.
2. Evaluate f at each critical number.
3. Evaluate $f(a)$ and $f(b)$.
4. The largest value is the absolute maximum; the smallest value is the absolute minimum.

Find the absolute extrema of

$$f(x) = 2x^3 - 3x^2 - 12x + 5 \quad \text{on } [-2, 4].$$

The critical numbers are -1 and 2 . Compare all candidates:

x	-2	-1	2	4
$f(x)$	1	12	-15	37

Hence

$$f(4) = 37 \text{ is the absolute maximum,} \quad f(2) = -15 \text{ is the absolute minimum.}$$

Find the absolute extrema of

$$f(x) = 4x^{5/4} - 8x^{1/4} \quad \text{on } [0, 4].$$

$$f'(x) = 5x^{1/4} - 2x^{-3/4} = \frac{5x - 2}{x^{3/4}}.$$

The candidates are $x = 0$, $x = \frac{2}{5}$, and $x = 4$.

$$f(0) = 0, \quad f\left(\frac{2}{5}\right) \approx -5.0897, \quad f(4) \approx 11.3137.$$

Therefore

$$f(4) \approx 11.3137 \text{ is the absolute maximum,}$$

$$f\left(\frac{2}{5}\right) \approx -5.0897 \text{ is the absolute minimum.}$$

Find the absolute extrema of

$$f(x) = x^3 - 5x + 3\sin(x^2) \quad \text{on } [-2, 2.5].$$

$$f'(x) = 3x^2 - 5 + 6x\cos(x^2).$$

Numerical root-finding gives four critical numbers:

$$a \approx -1.26410884789, \quad b \approx 0.674471354085,$$

$$c \approx 1.22668289470, \quad d \approx 2.01830371473.$$

Evaluating at these points and at the endpoints gives

x	a	b	c	d	-2	2.5
$f(x)$	7.3	-1.7	-1.3	-4.3	-0.3	3.0

Thus the absolute maximum is approximately 7.3 at $x \approx -1.2641$, and the absolute minimum is approximately -4.3 at $x \approx 2.0183$.

- The Extreme Value Theorem guarantees extrema when continuity and a closed bounded interval are present.
- Local extrema occur only at critical numbers, but not every critical number is an extremum.
- On a closed interval, compare endpoint values and values at all critical numbers.
- Numerical graphing and root-finding may be necessary when $f'(x) = 0$ cannot be solved algebraically.

Lesson 4.4: Increasing and Decreasing Functions

From the Sign of the Derivative to the Shape of the Graph

The derivative tells whether the function rises or falls. A sign chart for f' reveals intervals of monotonicity and classifies local extrema.

A function f is **increasing** on an interval I if $x_1 < x_2$ implies $f(x_1) < f(x_2)$. It is **decreasing** on I if $x_1 < x_2$ implies $f(x_1) > f(x_2)$.

Suppose f is differentiable on an interval I .

- (i) If $f'(x) > 0$ on I , then f is increasing on I .
- (ii) If $f'(x) < 0$ on I , then f is decreasing on I .

Proof using the Mean Value Theorem

For $x_1 < x_2$ in I , the Mean Value Theorem gives a $c \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c).$$

If $f'(c) > 0$, then $f(x_2) - f(x_1) > 0$, so $f(x_2) > f(x_1)$. The decreasing case is analogous.

For

$$f(x) = 2x^3 + 9x^2 - 24x - 10,$$

determine the intervals of increase and decrease and identify the local extrema.

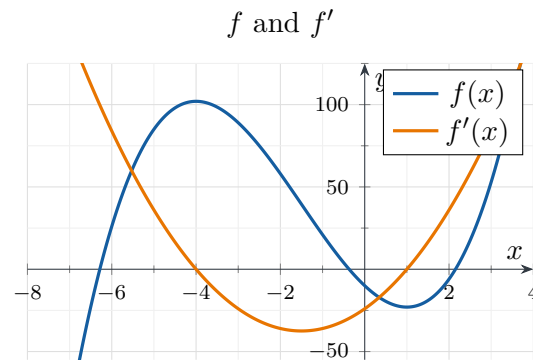
$$f'(x) = 6x^2 + 18x - 24 = 6(x - 1)(x + 4).$$

A sign chart gives

x	$(-\infty, -4)$	-4	$(-4, 1)$	1	$(1, \infty)$
$f'(x)$	+	0	-	0	+

Hence f increases on $(-\infty, -4) \cup (1, \infty)$ and decreases on $(-4, 1)$. Therefore

$(-4, 102)$ is a local maximum, $(1, -23)$ is a local minimum.



A default calculator window may hide local extrema or suggest false global behaviour. Calculus determines what features must exist; the graphing window should then be chosen to display them.

Graph

$$f(x) = 3x^4 + 40x^3 - 0.06x^2 - 1.2x$$

showing all local extrema.

$$f'(x) = 12(x - 0.1)(x + 0.1)(x + 10).$$

Thus

$$f'(x) < 0 \text{ on } (-\infty, -10) \cup (-0.1, 0.1),$$

$$f'(x) > 0 \text{ on } (-10, -0.1) \cup (0.1, \infty).$$

Therefore the graph has a local minimum at $x = -10$, a local maximum at $x = -0.1$, and a local minimum at $x = 0.1$. One wide window is needed for the global behaviour near $x = -10$; another close window is needed for the small-scale behaviour near the origin.

Let c be a critical number of a continuous function f .

- If f' changes from positive to negative at c , then $f(c)$ is a local maximum.
- If f' changes from negative to positive at c , then $f(c)$ is a local minimum.
- If f' does not change sign at c , then $f(c)$ is not a local extremum.

$$f' : + \rightarrow - \Rightarrow \text{local maximum}$$

$$f' : - \rightarrow + \Rightarrow \text{local minimum}$$

For $f(x) = 2x^3 + 9x^2 - 24x - 10$, f' changes $+$ $\rightarrow$ $-$ at $x = -4$, giving a local maximum, and changes $-$ $\rightarrow$ $+$ at $x = 1$, giving a local minimum.

Find the local extrema of

$$f(x) = x^{5/3} - 3x^{2/3}.$$

$$f'(x) = \frac{5x - 6}{3x^{1/3}}.$$

The critical numbers are $x = 0$ and $x = \frac{6}{5}$. The sign chart is

x	$(-\infty, 0)$	0	$(0, \frac{6}{5})$	$\frac{6}{5}$	$(\frac{6}{5}, \infty)$
$f'(x)$	+	DNE	-	0	+

Hence $x = 0$ is a local maximum and $x = \frac{6}{5}$ is a local minimum.

Find the local extrema of

$$f(x) = x^4 + 4x^3 - 5x^2 - 31x + 29.$$

$$f'(x) = 4x^3 + 12x^2 - 10x - 31.$$

Numerical root-finding gives

$$a \approx -2.96008, \quad b \approx -1.63816, \quad c \approx 1.59824.$$

The derivative is negative on $(-\infty, a) \cup (b, c)$ and positive on $(a, b) \cup (c, \infty)$. Therefore:

- local minimum at $x \approx -2.96008$;
- local maximum at $x \approx -1.63816$;
- local minimum at $x \approx 1.59824$ (also the absolute minimum).

- Positive derivative $\Rightarrow$ increasing; negative derivative $\Rightarrow$ decreasing.
- Classify critical numbers by the change in sign of f' .
- A critical number with no sign change is not a local extremum.
- Always compare calculus information with the graphing window; use both local and global views when necessary.

Lesson 4.5: Concavity and the Second Derivative Test

Beyond Rising and Falling

Two functions can both increase but bend in opposite ways. Concavity records whether tangent slopes are increasing or decreasing.

For a differentiable function f on an interval I :

- the graph is **concave up** when f' is increasing;
- the graph is **concave down** when f' is decreasing.

If f'' exists on an interval I :

- $f''(x) > 0$ on I implies the graph is concave up;
- $f''(x) < 0$ on I implies the graph is concave down.

If f is continuous at c and the graph changes concavity at c , then $(c, f(c))$ is an **inflection point**.

Solving $f''(x) = 0$ or finding where f'' is undefined produces only *candidates*. An inflection point exists only when the function is continuous there and the concavity actually changes.

For

$$f(x) = 2x^3 + 9x^2 - 24x - 10,$$

determine concavity and the inflection point.

$$f''(x) = 12x + 18.$$

Thus $f''(x) < 0$ for $x < -\frac{3}{2}$ and $f''(x) > 0$ for $x > -\frac{3}{2}$. Therefore the graph is concave down on $(-\infty, -\frac{3}{2})$, concave up on $(-\frac{3}{2}, \infty)$, and has an inflection point at

$$\left(-\frac{3}{2}, f\left(-\frac{3}{2}\right)\right) = \left(-\frac{3}{2}, \frac{79}{2}\right).$$

For

$$f(x) = x^4 - 6x^2 + 1,$$

find intervals of increase/decrease, concavity, extrema, and inflection points.

$$f'(x) = 4x(x - \sqrt{3})(x + \sqrt{3}), \quad f''(x) = 12(x - 1)(x + 1).$$

Hence

$$f' > 0 \quad \text{on } (-\sqrt{3}, 0) \cup (\sqrt{3}, \infty),$$

$$f' < 0 \quad \text{on } (-\infty, -\sqrt{3}) \cup (0, \sqrt{3}),$$

$$f'' > 0 \quad \text{on } (-\infty, -1) \cup (1, \infty),$$

$$f'' < 0 \quad \text{on } (-1, 1).$$

The graph has local minima at $x = \pm\sqrt{3}$, a local maximum at $x = 0$, and inflection points

$$(-1, -4) \text{ and } (1, -4).$$

For $f(x) = x^4$, $f''(x) = 12x^2$. Although $f''(0) = 0$, the second derivative is positive on both sides of 0. The graph is concave up throughout and has no inflection point.

Suppose $f'(c) = 0$ and f'' is continuous near c .

- If $f''(c) < 0$, then $f(c)$ is a local maximum.
- If $f''(c) > 0$, then $f(c)$ is a local minimum.

If $f''(c) = 0$ or $f''(c)$ does not exist, the Second Derivative Test gives no conclusion. Use the First Derivative Test or another argument.

Find the local extrema of

$$f(x) = x^4 - 8x^2 + 10.$$

$$f'(x) = 4x(x-2)(x+2), \quad f''(x) = 12x^2 - 16.$$

At the critical numbers:

$$f''(0) = -16 < 0 \Rightarrow \text{local maximum at } x = 0,$$

$$f''(\pm 2) = 32 > 0 \Rightarrow \text{local minima at } x = \pm 2.$$

The corresponding values are $f(0) = 10$ and $f(\pm 2) = -6$.

At the relevant critical point, the Second Derivative Test gives $f'' = 0$ for all three functions:

(a) $f(x) = x^3$: no local extremum at $x = 0$;

(b) $g(x) = (x+1)^4$: local minimum at $x = -1$;

(c) $h(x) = -x^4$: local maximum at $x = 0$.

This demonstrates why $f''(c) = 0$ is inconclusive.

Sketch

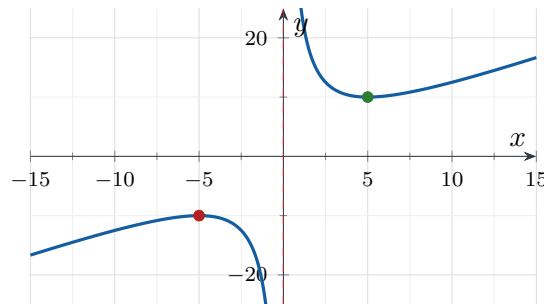
$$f(x) = x + \frac{25}{x}.$$

The domain is $\mathbb{R} \setminus \{0\}$.

$$f'(x) = \frac{(x-5)(x+5)}{x^2}, \quad f''(x) = \frac{50}{x^3}.$$

Thus f increases on $(-\infty, -5) \cup (5, \infty)$ and decreases on $(-5, 0) \cup (0, 5)$. There is a local maximum at $(-5, -10)$ and a local minimum at $(5, 10)$. The graph is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$. Since $x = 0$ is not in the domain, it is not an inflection point. The line $x = 0$ is a vertical asymptote.

$$f(x) = x + 25/x$$



Sketch

$$f(x) = (x+2)^{1/5} + 4.$$

The domain is all real numbers.

$$f'(x) = \frac{1}{5}(x+2)^{-4/5} > 0 \quad (x \neq -2),$$

so the function is increasing and has no local extrema. Also

$$f''(x) = -\frac{4}{25}(x+2)^{-9/5}.$$

This is positive on $(-\infty, -2)$ and negative on $(-2, \infty)$, so $(-2, 4)$ is an inflection point. Moreover $f'(x) \rightarrow +\infty$ from both sides as $x \rightarrow -2$, so the graph has a vertical tangent at $x = -2$.

- Concavity describes how tangent slopes change.
- Inflection points require continuity and a change in concavity.
- The Second Derivative Test is fast when $f''(c) \neq 0$.
- A discontinuity can separate concavity intervals but cannot itself be an inflection point.

Lesson 4.6: Overview of Curve Sketching

A Complete Calculus-Based Graphing Strategy

A representative graph is built from a systematic analysis, not from a default calculator window.

1. **Domain:** determine where f is defined.
2. **Discontinuities and vertical asymptotes:** evaluate one-sided limits at isolated excluded points.
3. **First derivative:** find critical numbers, intervals of increase/decrease, and local extrema.
4. **Vertical tangents:** at points in the domain where f' is undefined, evaluate one-sided limits of f' .
5. **Second derivative:** find concavity and inflection points.
6. **End behaviour:** compute limits as $x \rightarrow \pm\infty$ and identify horizontal or slant asymptotes.
7. **Intercepts:** find exact values where possible; otherwise approximate numerically.
8. **Window selection:** choose a scale that displays both local and global behaviour.

Sketch

$$f(x) = x^4 + 6x^3 + 12x^2 + 8x + 1.$$

The domain is $\mathbb{R}$, with no asymptotes.

$$f'(x) = 2(2x + 1)(x + 2)^2.$$

The critical numbers are $x = -2$ and $x = -\frac{1}{2}$. Since f' is negative on both sides of $x = -2$, there is no extremum there. At $x = -\frac{1}{2}$, f' changes from negative to positive, so there is a local minimum with

$$f\left(-\frac{1}{2}\right) = -\frac{11}{16}.$$

Also

$$f''(x) = 12(x + 2)(x + 1).$$

The graph is concave up on $(-\infty, -2) \cup (-1, \infty)$ and concave down on $(-2, -1)$. Inflection points occur at $(-2, 1)$ and $(-1, 0)$. The x -intercepts are $x = -1$ and $x \approx -0.160713$.

Sketch

$$f(x) = \frac{x^2 - 3}{x^3}.$$

The domain excludes $x = 0$, where there is a vertical asymptote:

$$\lim_{x \rightarrow 0^+} f(x) = -\infty, \quad \lim_{x \rightarrow 0^-} f(x) = +\infty.$$

$$f'(x) = \frac{9 - x^2}{x^4}.$$

Hence f increases on $(-3, 0) \cup (0, 3)$ and decreases on $(-\infty, -3) \cup (3, \infty)$. Thus $(-3, -2/9)$ is a local minimum and $(3, 2/9)$ is a local maximum.

$$f''(x) = \frac{2(x^2 - 18)}{x^5}.$$

Inflection points occur at $x = \pm 3\sqrt{2}$. The horizontal asymptote is $y = 0$, and the x -intercepts are $x = \pm\sqrt{3}$.

Sketch

$$f(x) = \frac{x^2}{x^2 - 4}.$$

The domain excludes $x = \pm 2$, both vertical asymptotes. Also

$$f'(x) = \frac{-8x}{(x^2 - 4)^2},$$

so f increases for $x < 0$ (within its domain) and decreases for $x > 0$. Therefore $(0, 0)$ is a local maximum.

$$f''(x) = \frac{8(3x^2 + 4)}{(x^2 - 4)^3}.$$

The graph is concave down on $(-2, 2)$ and concave up on $(-\infty, -2) \cup (2, \infty)$. There are no inflection points because ± 2 are not in the domain. The horizontal asymptote is $y = 1$.

Sketch

$$f(x) = \frac{1}{x^3 + 3x^2 + 3x + 3}.$$

The denominator is $(x+1)^3 + 2$, so it vanishes at

$$a = -1 - \sqrt[3]{2} \approx -2.259921.$$

Thus $x = a$ is a vertical asymptote. The derivative is

$$f'(x) = -\frac{3(x+1)^2}{(x^3 + 3x^2 + 3x + 3)^2} \leq 0,$$

so the function decreases on each part of its domain. The point $x = -1$ is a horizontal tangent but not an extremum. Further,

$$f''(x) = \frac{12x(x+1)(x^2 + 3x + 3)}{(x^3 + 3x^2 + 3x + 3)^3}.$$

The graph has inflection points at $(-1, \frac{1}{2})$ and $(0, \frac{1}{3})$. It has horizontal asymptote $y = 0$ and no x -intercepts.

Sketch

$$f(x) = e^{1/x}.$$

The domain is $(-\infty, 0) \cup (0, \infty)$. At $x = 0$,

$$\lim_{x \rightarrow 0^+} e^{1/x} = +\infty, \quad \lim_{x \rightarrow 0^-} e^{1/x} = 0.$$

Thus $x = 0$ is a one-sided vertical asymptote. Also

$$f'(x) = -\frac{e^{1/x}}{x^2} < 0,$$

so the function decreases on each domain interval.

$$f''(x) = e^{1/x} \frac{1 + 2x}{x^4}.$$

Hence the graph is concave down on $(-\infty, -\frac{1}{2})$ and concave up on $(-\frac{1}{2}, 0) \cup (0, \infty)$. The inflection point is

$$\left(-\frac{1}{2}, e^{-2}\right).$$

Finally, $y = 1$ is a horizontal asymptote as $x \rightarrow \pm\infty$, and there are no x -intercepts.

Sketch

$$f(x) = \cos x - x.$$

The domain is $\mathbb{R}$ and

$$f'(x) = -\sin x - 1 \leq 0.$$

Thus the function is decreasing everywhere. The derivative is zero at

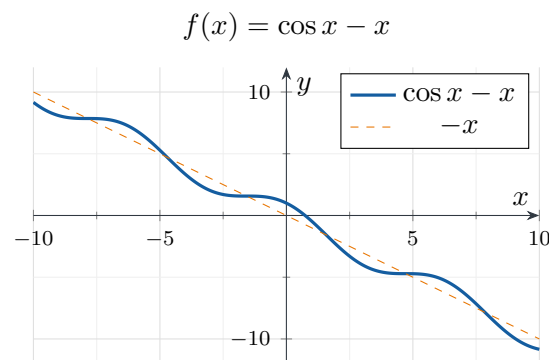
$$x = \frac{3\pi}{2} + 2n\pi, \quad n \in \mathbb{Z},$$

but it does not change sign, so there are no local extrema. Also

$$f''(x) = -\cos x,$$

so inflection points occur at odd multiples of $\pi/2$. Since $f(x) \rightarrow -\infty$ as $x \rightarrow \infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow -\infty$, there are no horizontal asymptotes. The unique x -intercept is

$$x \approx 0.739085.$$



A good graph is an argument supported by calculus. Domain and limits identify discontinuities and asymptotes; f' gives direction and extrema; f'' gives bending and inflection points; intercepts and end behaviour complete the picture. No single calculator window is guaranteed to reveal every feature.

Opening of Lesson 4.7: Optimization

Turning a Real Situation into a Maximum/Minimum Problem

Optimisation applies the extrema tools from Lessons 4.3–4.6 to real constraints.

1. Draw and label a diagram whenever possible.
2. Identify the variables and the relationship between them.
3. Decide which quantity must be maximised or minimised.
4. Use the constraint to express the objective in one variable.
5. Determine the physically allowable domain.

6. Find and compare all relevant candidate values.
7. Answer the original question with units and interpretation.

A rectangular garden is enclosed by 40 ft of fencing. Find the greatest possible area and the corresponding dimensions.

Let the dimensions be x and y . The perimeter constraint is

$$2x + 2y = 40 \implies y = 20 - x.$$

The area is

$$A(x) = xy = x(20 - x) = 20x - x^2, \quad 0 \leq x \leq 20.$$

$$A'(x) = 20 - 2x = 0 \implies x = 10.$$

Then $y = 10$ and

$$A_{\max} = 100 \text{ ft}^2, \quad 10 \text{ ft} \times 10 \text{ ft}.$$

The maximum occurs for a square.

Integrated Reference Tables

Information	Positive	Negative
$f'(x)$	f is increasing	f is decreasing
Sign change of f'	$+$ $\rightarrow$ $-$: local maximum	$-$ $\rightarrow$ $+$: local minimum
$f''(x)$	graph is concave up	graph is concave down
At $f'(c) = 0$	$f''(c) > 0$: local minimum	$f''(c) < 0$: local maximum

1. $f'(c) = 0$ does not guarantee an extremum.
2. f' undefined at c does not guarantee an extremum.
3. $f''(c) = 0$ does not guarantee an inflection point.
4. A change in concavity across an excluded point does not make that point an inflection point.

1. Find the domain.
2. Solve $f'(x) = 0$ and include domain points where f' is undefined.
3. Build the f' sign chart.
4. Solve $f''(x) = 0$ and include points where f'' is undefined.

5. Build the f'' sign chart and verify continuity before declaring inflection points.
6. Add limits, asymptotes, and intercepts.

Textbook Exercise Coverage Map

Assigned Pages and Exercise Sets

PDF pages	Textbook pages	Included content
40–50	250–260	Lesson 4.3 definitions, theorems, proofs, Examples 3.1–3.13, writing exercises, full exercise set, applications, and exploratory exercises; the page also transitions into Lesson 4.4.
51–58	261–268	Lesson 4.4 definitions, theorem proof, Examples 4.1–4.5, First Derivative Test, writing exercises, full exercise set, applications, and exploratory exercises.
59–67	269–277	Lesson 4.5 definitions, Examples 5.1–5.7, Second Derivative Test, writing exercises, full exercise set, applications, and exploratory exercises.
68–78	278–288	Lesson 4.6 complete curve-sketching procedure, Examples 6.1–6.6, writing exercises, full exercise set, applications, exploratory exercises, and the opening of Lesson 4.7 with Example 7.1.

Complete Assigned Textbook Pages 40–78

The following 39 pages are included exactly from the uploaded textbook so that every original question, graph, table, application, historical note, exploratory exercise, and diagram in the assigned range is preserved. This appendix is especially useful for assigning the original numbered exercises alongside the retypeset notes and worked examples in the main body.

Maximum and Minimum Values

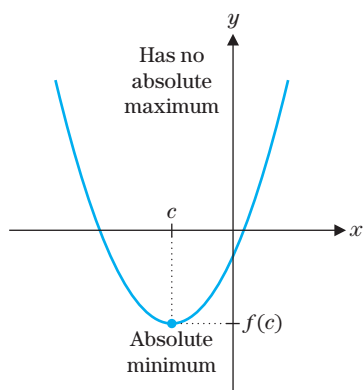


FIGURE 4.24a

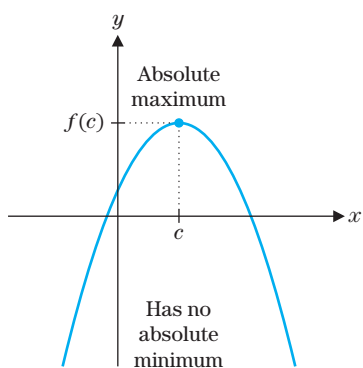


FIGURE 4.24b

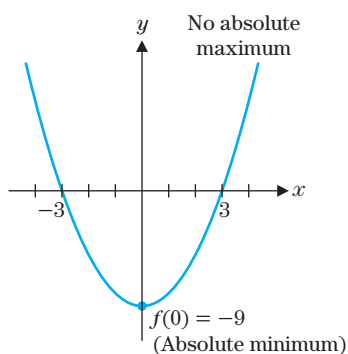


FIGURE 4.25

$$y = x^2 - 9 \text{ on } (-\infty, \infty)$$

To remain competitive, businesses must regularly evaluate how to minimize waste and maximize the return on their investment. In this section, we consider the problem of finding maximum and minimum values of functions. In section 4.7, we examine how to apply these notions to problems of an applied nature.

We begin by giving careful mathematical definitions of some familiar terms.

DEFINITION 3.1

For a function f defined on a set S of real numbers and a number $c \in S$,

- (i) $f(c)$ is the **absolute maximum** of f on S if $f(c) \geq f(x)$ for all $x \in S$ and
- (ii) $f(c)$ is the **absolute minimum** of f on S if $f(c) \leq f(x)$ for all $x \in S$.

An absolute maximum or an absolute minimum is referred to as an **absolute extremum**. (The plural form of extremum is **extrema**.)

The first question you might ask is whether every function has an absolute maximum and an absolute minimum. The answer is no, as we can see from Figures 4.24a and 4.24b.

EXAMPLE 3.1 Absolute Maximum and Minimum Values

- (a) Locate any absolute extrema of $f(x) = x^2 - 9$ on the interval $(-\infty, \infty)$. (b) Locate any absolute extrema of $f(x) = x^2 - 9$ on the interval $(-3, 3)$. (c) Locate any absolute extrema of $f(x) = x^2 - 9$ on the interval $[-3, 3]$.

Solution (a) In Figure 4.25, notice that f has an absolute minimum value of $f(0) = -9$, but has no absolute maximum value.

(b) In Figure 4.26a, we see that f has an absolute minimum value of $f(0) = -9$. Your initial reaction might be to say that f has an absolute maximum of 0, but $f(x) \neq 0$ for any $x \in (-3, 3)$, since this is an open interval. Hence, f has no absolute maximum on the interval $(-3, 3)$.

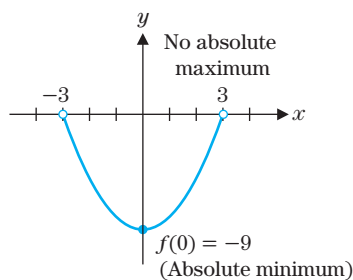


FIGURE 4.26a

$$y = x^2 - 9 \text{ on } (-3, 3)$$

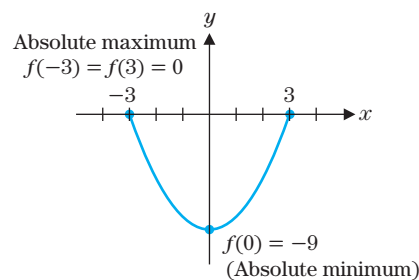


FIGURE 4.26b

$$y = x^2 - 9 \text{ on } [-3, 3]$$

- (c) In this case, the endpoints 3 and -3 are in the interval $[-3, 3]$. Here, f assumes its absolute maximum at two points: $f(3) = f(-3) = 0$. (See Figure 4.26b.) ■

We have seen that a function may or may not have absolute extrema. In example 3.1, the function failed to have an absolute maximum, except on the closed, bounded interval $[-3, 3]$. Example 3.2 provides another piece of the puzzle.

EXAMPLE 3.2 A Function with No Absolute Maximum or Minimum

Locate any absolute extrema of $f(x) = 1/x$, on $[-3, 0) \cup (0, 3]$.

Solution From the graph in Figure 4.27, f clearly fails to have either an absolute maximum or an absolute minimum on $[-3, 0) \cup (0, 3]$. The following table of values for $f(x)$ for x close to 0 suggests the same conclusion.

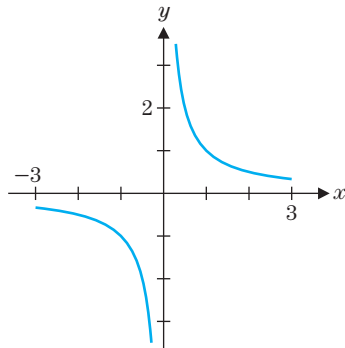


FIGURE 4.27
 $y = 1/x$

x	$1/x$
1	1
0.1	10
0.01	100
0.001	1000
0.0001	10,000
0.00001	100,000
0.000001	1,000,000

x	$1/x$
-1	-1
-0.1	-10
-0.01	-100
-0.001	-1000
-0.0001	-10,000
-0.00001	-100,000
-0.000001	-1,000,000

The most obvious difference between the functions in examples 3.1 and 3.2 is that $f(x) = 1/x$ is not continuous throughout the interval $[-3, 3]$. We offer the following theorem without proof.

THEOREM 3.1 (Extreme Value Theorem)

A continuous function f defined on a *closed, bounded* interval $[a, b]$ attains both an absolute maximum and an absolute minimum on that interval.

While you do not need to have a continuous function or a closed interval to have an absolute extremum, Theorem 3.1 says that continuous functions are *guaranteed* to have an absolute maximum and an absolute minimum on a closed, bounded interval.

In example 3.3, we revisit the function from example 3.2, but look on a different interval.

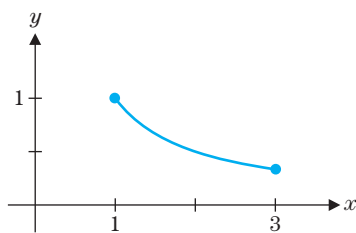


FIGURE 4.28
 $y = 1/x$ on $[1, 3]$

EXAMPLE 3.3 Finding Absolute Extrema of a Continuous Function

Find the absolute extrema of $f(x) = 1/x$ on the interval $[1, 3]$.

Solution Notice that on the interval $[1, 3]$, f is continuous. Consequently, the Extreme Value Theorem guarantees that f has both an absolute maximum and an absolute minimum on $[1, 3]$. Judging from the graph in Figure 4.28, it appears that $f(x)$ reaches its maximum value of 1 at $x = 1$ and its minimum value of $1/3$ at $x = 3$.

Our objective is to determine how to locate the absolute extrema of a given function. Before we do this, we need to consider an additional type of extremum.

DEFINITION 3.2

- (i) $f(c)$ is a **local maximum** of f if $f(c) \geq f(x)$ for all x in some *open* interval containing c .
- (ii) $f(c)$ is a **local minimum** of f if $f(c) \leq f(x)$ for all x in some *open* interval containing c .

In either case, we call $f(c)$ a **local extremum** of f .

REMARK 3.1

Local maxima and minima (the plural forms of maximum and minimum, respectively) are sometimes referred to as **relative** maxima and minima, respectively.

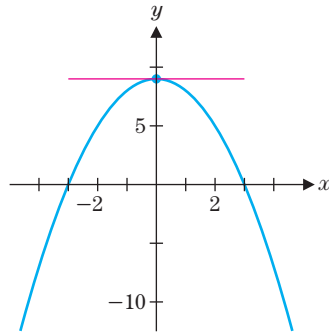


FIGURE 4.30
 $y = 9 - x^2$ and the tangent line at $x = 0$

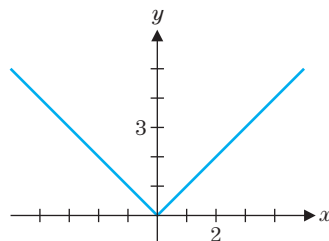


FIGURE 4.31
 $y = |x|$



HISTORICAL NOTES

Pierre de Fermat (1601–1665) A French mathematician who discovered many important results, including the theorem named for him. Fermat was a lawyer and member of the Toulouse supreme court, with mathematics as a hobby. The “Prince of Amateurs” left an unusual legacy by writing in the margin of a book that he had discovered a wonderful proof of a clever result, but that the margin of the book was too small to hold the proof. Fermat’s Last Theorem confounded many of the world’s best mathematicians for more than 300 years before being proved by Andrew Wiles in 1995.

Notice from Figure 4.29 that each local extremum seems to occur either at a point where the tangent line is horizontal [i.e., where $f'(x) = 0$], at a point where the tangent line is vertical [where $f'(x)$ is undefined] or at a corner [again, where $f'(x)$ is undefined]. We can see this behavior quite clearly in examples 3.4 and 3.5.

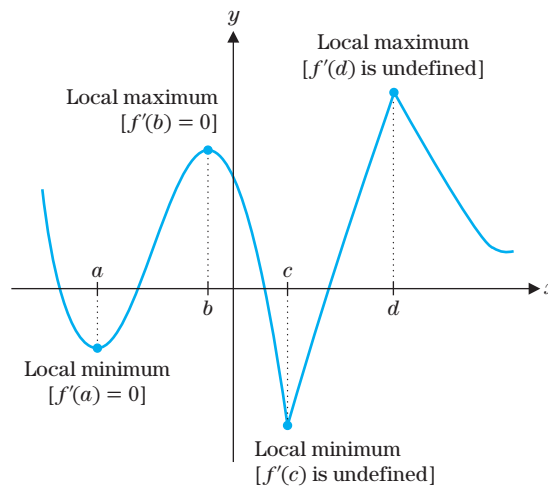


FIGURE 4.29
Local extrema

EXAMPLE 3.4 A Function with a Zero Derivative at a Local Maximum

Locate any local extrema for $f(x) = 9 - x^2$ and describe the behavior of the derivative at the local extremum.

Solution We can see from Figure 4.30 that there is a local maximum at $x = 0$. Further, note that $f'(x) = -2x$ and so, $f'(0) = 0$. This says that the tangent line to $y = f(x)$ at $x = 0$ is horizontal, as indicated in Figure 4.30. ■

EXAMPLE 3.5 A Function with an Undefined Derivative at a Local Minimum

Locate any local extrema for $f(x) = |x|$ and describe the behavior of the derivative at the local extremum.

Solution We can see from Figure 4.31 that there is a local minimum at $x = 0$. The graph has a corner at $x = 0$ and hence, $f'(0)$ is undefined. ■

The graphs shown in Figures 4.29–4.31 are not unusual. In fact, spend a little time now drawing graphs of functions with local extrema. It should not take long to convince yourself that local extrema occur only at points where the derivative is either zero or undefined. Because of this, we give these points a special name.

DEFINITION 3.3

A number c in the domain of a function f is called a **critical number** of f if $f'(c) = 0$ or $f'(c)$ is undefined.

It turns out that our earlier observation regarding the location of extrema is correct. That is, local extrema occur *only* at points where the derivative is zero or undefined. We state this formally in Theorem 3.2.

THEOREM 3.2 (Fermat's Theorem)

Suppose that $f(c)$ is a local extremum (local maximum or local minimum). Then c must be a critical number of f .

PROOF

Suppose that f is differentiable at $x = c$. (If not, c is a critical number of f and we are done.) Suppose further that $f'(c) \neq 0$. Then, either $f'(c) > 0$ or $f'(c) < 0$.

If $f'(c) > 0$, we have by the definition of derivative that

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} > 0.$$

So, for all h sufficiently small,

$$\frac{f(c+h) - f(c)}{h} > 0. \quad (3.1)$$

For $h > 0$, (3.1) says that

$$f(c+h) - f(c) > 0$$

and so,

$$f(c+h) > f(c).$$

Thus, $f(c)$ is not a local maximum.

Similarly, for $h < 0$, (3.1) says that

$$f(c+h) - f(c) < 0$$

and so,

$$f(c+h) < f(c).$$

Thus, $f(c)$ is not a local minimum, either.

Since we had assumed that $f(c)$ was a local extremum, this is a contradiction. This rules out the possibility that $f'(c) > 0$.

We leave it as an exercise to show that if $f'(c) < 0$, we obtain the same contradiction. The only remaining possibility is to have $f'(c) = 0$ and this proves the theorem. ■

We can use Fermat's Theorem and calculator- or computer-generated graphs to find local extrema, as in examples 3.6 and 3.7.

EXAMPLE 3.6 Finding Local Extrema of a Polynomial

Find the critical numbers and local extrema of $f(x) = 2x^3 - 3x^2 - 12x + 5$.

Solution Here,

$$\begin{aligned} f'(x) &= 6x^2 - 6x - 12 = 6(x^2 - x - 2) \\ &= 6(x-2)(x+1). \end{aligned}$$



TODAY IN MATHEMATICS

Andrew Wiles (1953–)
A British mathematician who in 1995 published a proof of Fermat's Last Theorem, the most famous unsolved problem of the 20th century. Fermat's Last Theorem states that there is no integer solution x, y and z of the equation $x^n + y^n = z^n$ for integers $n > 2$. Wiles had wanted to prove the theorem since reading about it as a 10-year-old. After more than ten years as a successful research mathematician, Wiles isolated himself from colleagues for seven years as he developed the mathematics needed for his proof. "I realised that talking to people casually about Fermat was impossible because it generated too much interest. You cannot focus yourself for years unless you have this kind of undivided concentration which too many spectators would destroy." The last step of his proof came, after a year of intense work on this one step, as "this incredible revelation" that was "so indescribably beautiful, it was so simple and elegant."

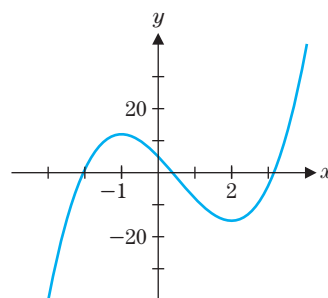


FIGURE 4.32
 $y = 2x^3 - 3x^2 - 12x + 5$

Thus, f has two critical numbers, $x = -1$ and $x = 2$. Notice from the graph in Figure 4.32 that these correspond to the locations of a local maximum and a local minimum, respectively. ■

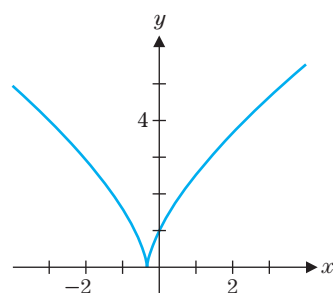


FIGURE 4.33
 $y = (3x + 1)^{2/3}$

EXAMPLE 3.7 An Extremum at a Point Where the Derivative Is Undefined

Find the critical numbers and local extrema of $f(x) = (3x + 1)^{2/3}$.

Solution Here, we have

$$f'(x) = \frac{2}{3}(3x + 1)^{-1/3}(3) = \frac{2}{(3x + 1)^{1/3}}.$$

Of course, $f'(x) \neq 0$ for all x , but $f'(x)$ is undefined at $x = -\frac{1}{3}$. Be sure to note that $-\frac{1}{3}$ is in the domain of f . Thus, $x = -\frac{1}{3}$ is the only critical number of f . From the graph in Figure 4.33, we see that this corresponds to the location of a local minimum (also the absolute minimum). If you use your graphing utility to try to produce a graph of $y = f(x)$, you may get only half of the graph displayed in Figure 4.33. The reason is that the algorithms used by most calculators and many computers will return a complex number (or an error) when asked to compute certain fractional powers of negative numbers. While this annoying shortcoming presents only occasional difficulties, we mention this here only so that you are aware that technology has limitations. ■

REMARK 3.2

Fermat's Theorem says that local extrema can occur only at critical numbers. This does not say that there is a local extremum at *every* critical number. In fact, this is false, as we illustrate in examples 3.8 and 3.9.

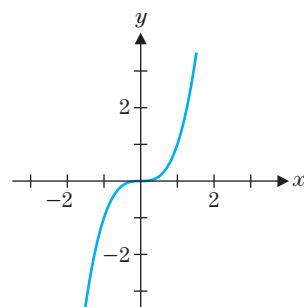


FIGURE 4.34
 $y = x^3$

EXAMPLE 3.8 A Horizontal Tangent at a Point That Is Not a Local Extremum

Find the critical numbers and local extrema of $f(x) = x^3$.

Solution It should be clear from Figure 4.34 that f has no local extrema. However, $f'(x) = 3x^2 = 0$ for $x = 0$ (the only critical number of f). In this case, f has a horizontal tangent line at $x = 0$, but does not have a local extremum there. ■

EXAMPLE 3.9 A Vertical Tangent at a Point That Is Not a Local Extremum

Find the critical numbers and local extrema of $f(x) = x^{1/3}$.

Solution As in example 3.8, f has no local extrema. (See Figure 4.35 on the following page.) Here, $f'(x) = \frac{1}{3}x^{-2/3}$ and so, f has a critical number at $x = 0$. (In this case, the derivative is undefined at $x = 0$.) However, f does not have a local extremum at $x = 0$. ■

You should always check that a given value is in the domain of the function before declaring it a critical number, as in example 3.10.

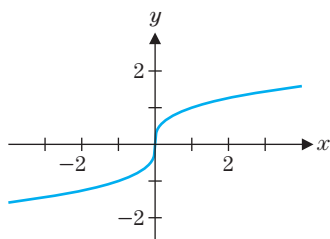


FIGURE 4.35
 $y = x^{1/3}$

REMARK 3.3

When we use the terms maximum, minimum or extremum without specifying absolute or local, we will *always* be referring to absolute extrema.

EXAMPLE 3.10 Finding Critical Numbers of a Rational Function

Find all the critical numbers of $f(x) = \frac{2x^2}{x+2}$.

Solution You should note that the domain of f consists of all real numbers other than $x = -2$. Here, we have

$$\begin{aligned} f'(x) &= \frac{4x(x+2) - 2x^2(1)}{(x+2)^2} && \text{From the quotient rule.} \\ &= \frac{2x(x+4)}{(x+2)^2}. \end{aligned}$$

Notice that $f'(x) = 0$ for $x = 0, -4$ and $f'(x)$ is undefined for $x = -2$. However, -2 is not in the domain of f and consequently, the only critical numbers are $x = 0$ and $x = -4$. ■

We have observed that local extrema occur only at critical numbers and that continuous functions must have an absolute maximum and an absolute minimum on a closed, bounded interval. Theorem 3.3 gives us a way to find absolute extrema.

THEOREM 3.3

Suppose that f is continuous on the closed interval $[a, b]$. Then, each absolute extremum of f must occur at an endpoint (a or b) or at a critical number.

PROOF

By the Extreme Value Theorem, f will attain its maximum and minimum values on $[a, b]$, since f is continuous. Let $f(c)$ be an absolute extremum. If c is *not* an endpoint (i.e., $c \neq a$ and $c \neq b$), then c must be in the open interval (a, b) . In this case, $f(c)$ is also a local extremum. By Fermat's Theorem, then, c must be a critical number, since local extrema occur only at critical numbers. ■

REMARK 3.4

Theorem 3.3 gives us a simple procedure for finding the absolute extrema of a continuous function on a closed, bounded interval:

1. Find all critical numbers in the interval and compute function values at these points.
2. Compute function values at the endpoints.
3. The largest of these function values is the absolute maximum and the smallest of these function values is the absolute minimum.

We illustrate Theorem 3.3 for the case of a polynomial function in example 3.11.

EXAMPLE 3.11 Finding Absolute Extrema on a Closed Interval

Find the absolute extrema of $f(x) = 2x^3 - 3x^2 - 12x + 5$ on the interval $[-2, 4]$.

Solution From the graph in Figure 4.36, the maximum appears to be at the endpoint $x = 4$, while the minimum appears to be at a local minimum near $x = 2$. In example 3.6, we found that the critical numbers of f are $x = -1$ and $x = 2$. Further, both of these are in the interval $[-2, 4]$. So, we compare the values at the endpoints:

$$f(-2) = 1 \quad \text{and} \quad f(4) = 37,$$

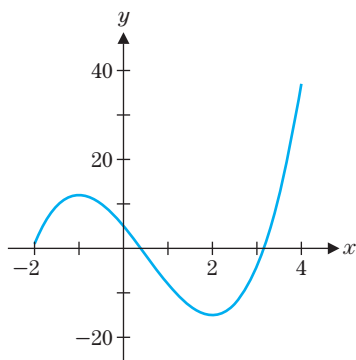


FIGURE 4.36
 $y = 2x^3 - 3x^2 - 12x + 5$

and the values at the critical numbers:

$$f(-1) = 12 \quad \text{and} \quad f(2) = -15.$$

Since f is continuous on $[-2, 4]$, Theorem 3.3 says that the absolute extrema must be among these four values. Thus, $f(4) = 37$ is the absolute maximum and $f(2) = -15$ is the absolute minimum, which is consistent with what we see in the graph in Figure 4.36. ■

Of course, most real problems of interest are unlikely to result in derivatives with integer zeros. Consider the following somewhat less user-friendly example.

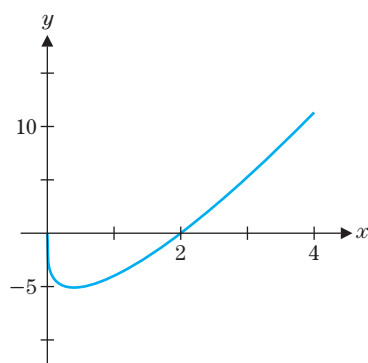


FIGURE 4.37
 $y = 4x^{5/4} - 8x^{1/4}$

EXAMPLE 3.12 Finding Extrema for a Function with Fractional Exponents

Find the absolute extrema of $f(x) = 4x^{5/4} - 8x^{1/4}$ on the interval $[0, 4]$.

Solution From the graph in Figure 4.37, it appears that the maximum occurs at the endpoint $x = 4$ and the minimum near $x = \frac{1}{2}$. Next, observe that

$$f'(x) = 5x^{1/4} - 2x^{-3/4} = \frac{5x - 2}{x^{3/4}}.$$

Thus, the critical numbers are $x = \frac{2}{5}$ [since $f'(\frac{2}{5}) = 0$] and $x = 0$ (since $f'(0)$ is undefined and 0 is in the domain of f). We now need only compare

$$f(0) = 0, \quad f(4) \approx 11.3137 \quad \text{and} \quad f\left(\frac{2}{5}\right) \approx -5.0897.$$

So, the absolute maximum is $f(4) \approx 11.3137$ and the absolute minimum is $f(\frac{2}{5}) \approx -5.0897$, which is consistent with what we expected from Figure 4.37. ■

In practice, the critical numbers are not always as easy to find as they were in examples 3.11 and 3.12. In example 3.13, it is not even known *how many* critical numbers there are. We can, however, estimate the number and locations of these from a careful analysis of computer-generated graphs.

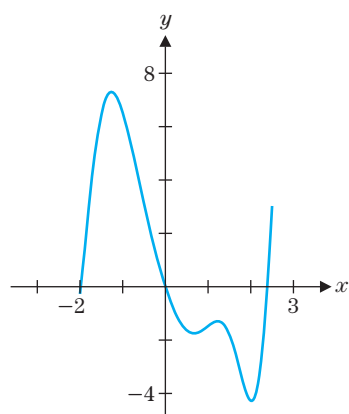


FIGURE 4.38
 $y = f(x) = x^3 - 5x + 3 \sin x^2$

EXAMPLE 3.13 Finding Absolute Extrema Approximately

Find the absolute extrema of $f(x) = x^3 - 5x + 3 \sin x^2$ on the interval $[-2, 2.5]$.

Solution From the graph in Figure 4.38, it appears that the maximum occurs near $x = -1$, while the minimum seems to occur near $x = 2$. Next, we compute

$$f'(x) = 3x^2 - 5 + 6x \cos x^2.$$

Unlike examples 3.11 and 3.12, there is no algebra we can use to find the zeros of f' . Our only alternative is to find the zeros approximately. You can do this by using Newton's method to solve $f'(x) = 0$. (You can also use any other rootfinding method built into your calculator or computer.) First, we'll need adequate initial guesses. From the graph of $y = f'(x)$ found in Figure 4.39, it appears that there are four zeros of $f'(x)$ on the interval in question, located near $x = -1.3, 0.7, 1.2$ and 2.0 . Further, referring back to Figure 4.38, these four zeros correspond with the four local extrema seen in the graph of $y = f(x)$. We now apply Newton's method to solve $f'(x) = 0$, using the preceding four values as our initial guesses. This leads us to four approximate critical numbers of f on the interval $[-2, 2.5]$. We have

$$a \approx -1.26410884789, \quad b \approx 0.674471354085,$$

$$c \approx 1.2266828947 \quad \text{and} \quad d \approx 2.01830371473.$$

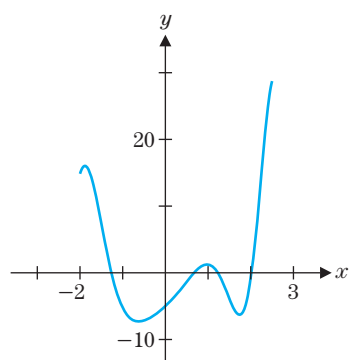


FIGURE 4.39
 $y = f'(x) = 3x^2 - 5 + 6x \cos x^2$

We now need only compare the values of f at the endpoints and the approximate critical numbers:

$$\begin{aligned} f(a) &\approx 7.3, & f(b) &\approx -1.7, & f(c) &\approx -1.3 \\ f(d) &\approx -4.3, & f(-2) &\approx -0.3 & \text{and} & f(2.5) \approx 3.0. \end{aligned}$$

Thus, the absolute maximum is approximately $f(-1.26410884789) \approx 7.3$ and the absolute minimum is approximately $f(2.01830371473) \approx -4.3$.

It is important (especially in light of how much of our work here was approximate and graphical) to verify that the approximate extrema correspond with what we expect from the graph of $y = f(x)$. Since these correspond closely, we have great confidence in their accuracy. ■

We have now seen how to locate the absolute extrema of a continuous function on a closed interval. In section 4.4, we see how to find local extrema.

BEYOND FORMULAS

The Extreme Value Theorem is an important but subtle result. Think of it this way. If the hypotheses of the theorem are met, you will never waste your time looking for the maximum of a function that does not have a maximum. That is, the problem is always solvable. The technique described in Remark 3.4 *always works*, as long as there are only finitely many critical numbers.

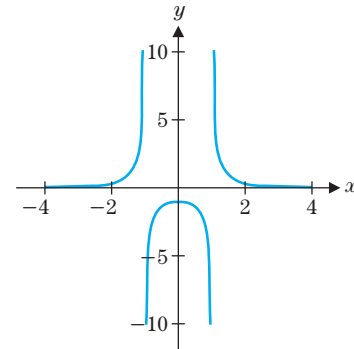
EXERCISES 4.3

WRITING EXERCISES

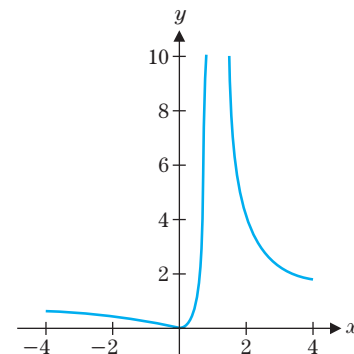
- Using one or more graphs, explain why the Extreme Value Theorem is true. Is the conclusion true if we drop the hypothesis that f is a continuous function? Is the conclusion true if we drop the hypothesis that the interval is closed?
- Using one or more graphs, argue that Fermat's Theorem is true. Discuss how Fermat's Theorem is used. Restate the theorem in your own words to make its use clearer.
- Suppose that $f(t)$ represents your elevation after t hours on a mountain hike. If you stop to rest, explain why $f'(t) = 0$. Discuss the circumstances under which you would be at a local maximum, local minimum or neither.
- Mathematically, an if/then statement is usually strictly one-directional. When we say "If A , then B " it is generally *not* the case that "If B , then A " is also true. When both are true, we say " A if and only if B ," which is abbreviated to " A iff B ." Consider the statement, "If you stood in the rain, *then* you got wet." Is this true? How does this differ from its **converse**, "If you got wet, *then* you stood in the rain." Apply this logic to both the Extreme Value Theorem and Fermat's Theorem: state the converse and decide whether its conclusion is sometimes true or always true.

In exercises 1 and 2, use the graph to locate the absolute extrema (if they exist) of the function on the given interval.

- $f(x) = \frac{1}{x^2 - 1}$ on (a) $(0, 1) \cup (1, \infty)$,
(b) $(-1, 1)$, (c) $(0, 1)$, (d) $\left[-\frac{1}{2}, \frac{1}{2}\right]$



- $f(x) = \frac{x^2}{(x-1)^2}$ on (a) $(-\infty, 1) \cup (1, \infty)$,
(b) $(-1, 1)$, (c) $(0, 1)$, (d) $[-2, -1]$



In exercises 3–6, find all critical numbers by hand. Use your knowledge of the type of graph (e.g., parabola or cubic) to determine whether the critical number represents a local maximum, local minimum or neither.

3. (a) $f(x) = x^2 + 5x - 1$ (b) $f(x) = -x^2 + 4x + 2$
4. (a) $f(x) = x^3 - 3x + 1$ (b) $f(x) = -x^3 + 6x^2 + 2$
5. (a) $f(x) = x^3 - 3x^2 + 6x$ (b) $f(x) = -x^3 + 3x^2 - 3x$
6. (a) $f(x) = x^4 - 2x^2 + 1$ (b) $f(x) = x^4 - 3x^3 + 2$

In exercises 7–24, find all critical numbers by hand. If available, use graphing technology to determine whether the critical number represents a local maximum, local minimum or neither.

7. $f(x) = x^4 - 3x^3 + 2$ 8. $f(x) = x^4 + 6x^2 - 2$
9. $f(x) = x^{3/4} - 4x^{1/4}$ 10. $f(x) = (x^{2/5} - 3x^{1/5})^2$
11. $f(x) = \sin x \cos x, [0, 2\pi]$ 12. $f(x) = \sqrt{3} \sin x + \cos x$
13. $f(x) = \frac{x^2 - 2}{x + 2}$ 14. $f(x) = \frac{x^2 - x + 4}{x - 1}$
15. $f(x) = \frac{1}{2}(e^x + e^{-x})$ 16. $f(x) = xe^{-2x}$
17. $f(x) = x^{4/3} + 4x^{1/3} + 4x^{-2/3}$ 18. $f(x) = x^{7/3} - 28x^{1/3}$
19. $f(x) = 2x\sqrt{x+1}$ 20. $f(x) = x/\sqrt{x^2+1}$
21. $f(x) = |x^2 - 1|$ 22. $f(x) = \sqrt[3]{x^3 - 3x^2}$
23. $f(x) = \begin{cases} x^2 + 2x - 1 & \text{if } x < 0 \\ x^2 - 4x + 3 & \text{if } x \geq 0 \end{cases}$
24. $f(x) = \begin{cases} \sin x & \text{if } -\pi < x < \pi \\ -\tan x & \text{if } |x| \geq \pi \end{cases}$

In exercises 25–34, find the absolute extrema of the given function on each indicated interval.

25. $f(x) = x^3 - 3x + 1$ on (a) $[0, 2]$ and (b) $[-3, 2]$
26. $f(x) = x^4 - 8x^2 + 2$ on (a) $[-3, 1]$ and (b) $[-1, 3]$
27. $f(x) = x^{2/3}$ on (a) $[-4, -2]$ and (b) $[-1, 3]$
28. $f(x) = \sin x + \cos x$ on (a) $[0, 2\pi]$ and (b) $[\pi/2, \pi]$
29. $f(x) = e^{-x^2}$ on (a) $[0, 2]$ and (b) $[-3, 2]$
30. $f(x) = x^2 e^{-4x}$ on (a) $[-2, 0]$ and (b) $[0, 4]$
31. $f(x) = \frac{3x^2}{x-3}$ on (a) $[-2, 2]$ and (b) $[2, 8]$
32. $f(x) = \tan^{-1}(x^2)$ on (a) $[0, 1]$ and (b) $[-3, 4]$
33. $f(x) = \frac{x}{x^2 + 1}$ on (a) $[0, 2]$ and (b) $[-3, 3]$
34. $f(x) = \frac{3x}{x^2 + 16}$ on (a) $[0, 2]$ and (b) $[0, 6]$

In exercises 35–38, numerically estimate the absolute extrema of the given function on the indicated intervals.

35. $f(x) = x^4 - 3x^2 + 2x + 1$ on (a) $[-1, 1]$ and (b) $[-3, 2]$
36. $f(x) = x^6 - 3x^4 - 2x + 1$ on (a) $[-1, 1]$ and (b) $[-2, 2]$
37. $f(x) = x \sin x + 3$ on (a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and (b) $[0, 2\pi]$
38. $f(x) = x^2 + e^x$ on (a) $[0, 1]$ and (b) $[-2, 2]$

39. Sketch a graph of a function f such that the absolute maximum of $f(x)$ on the interval $[-2, 2]$ equals 3 and the absolute minimum does not exist.
40. Sketch a graph of a continuous function f such that the absolute maximum of $f(x)$ on the interval $(-2, 2)$ does not exist and the absolute minimum equals 2.
41. Sketch a graph of a continuous function f such that the absolute maximum of $f(x)$ on the interval $(-2, 2)$ equals 4 and the absolute minimum equals 2.
42. Sketch a graph of a function f such that the absolute maximum of $f(x)$ on the interval $[-2, 2]$ does not exist and the absolute minimum does not exist.
43. In this exercise, we will explore the family of functions $f(x) = x^3 + cx + 1$, where c is constant. How many and what types of local extrema are there? (Your answer will depend on the value of c .) Assuming that this family is indicative of all cubic functions, list all types of cubic functions.
44. Prove that any fourth-order polynomial must have at least one local extremum and can have a maximum of three local extrema. Based on this information, sketch several possible graphs of fourth-order polynomials.
45. Show that $f(x) = x^3 + bx^2 + cx + d$ has both a local maximum and a local minimum if $c < 0$.
46. In exercise 45, show that the sum of the critical numbers is $-\frac{2b}{3}$.
47. For the family of functions $f(x) = x^4 + cx^2 + 1$, find all local extrema (your answer will depend on the value of the constant c).
48. For the family of functions $f(x) = x^4 + cx^3 + 1$, find all local extrema. (Your answer will depend on the value of the constant c .)
49. If f is differentiable on the interval $[a, b]$ and $f'(a) < 0 < f'(b)$, prove that there is a c with $a < c < b$ for which $f'(c) = 0$. (Hint: Use the Extreme Value Theorem and Fermat's Theorem.)
50. Sketch a graph showing that $y = f(x) = x^2 + 1$ and $y = g(x) = \ln x$ do not intersect. Find x to minimize $f(x) - g(x)$. At this value of x , show that the tangent lines to $y = f(x)$ and $y = g(x)$ are parallel. Explain graphically why it makes sense that the tangent lines are parallel.
51. Sketch a graph of $f(x) = \frac{x^2}{x^2 + 1}$ for $x > 0$ and determine where the graph is steepest. (That is, find where the slope is a maximum.)

52. Sketch a graph of $f(x) = e^{-x^2}$ and determine where the graph is steepest. (Note: This is an important problem in probability theory.)



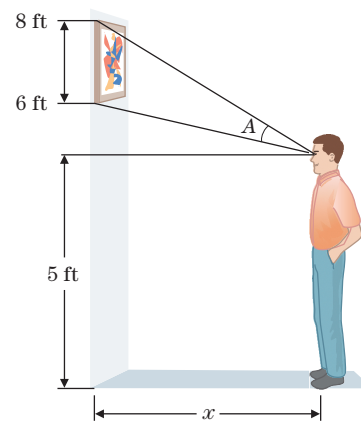
APPLICATIONS

53. If two soccer teams each score goals at a rate of r goals per minute, the probability that n goals will be scored in t minutes is $P = \frac{(rt)^n}{n!} e^{-rt}$. Take $r = \frac{1}{25}$. Show that for a 90 minute game, P is maximized with $n = 3$. Briefly explain why this makes sense. Find t to maximize the probability that exactly 1 goal has been scored. Briefly explain why your answer makes sense.
54. If you have won three out of four matches against someone, does that mean that the probability that you will win the next one is $\frac{3}{4}$? In general, if you have a probability p of winning each match, the probability of winning m out of n matches is $f(p) = \frac{n!}{(n-m)!m!} p^m (1-p)^{n-m}$. Find p to maximize f . This value of p is called the **maximum likelihood estimator** of the probability. Briefly explain why your answer makes sense.



55. A section of roller coaster is in the shape of $y = x^5 - 4x^3 - x + 10$, where x is between -2 and 2 . Find all local extrema and explain what portions of the roller coaster they represent. Find the location of the steepest part of the roller coaster.
56. Suppose a large computer file is sent over the Internet. If the probability that it reaches its destination without any errors is x , then the probability that an error is made is $1 - x$. The field of information theory studies such situations. An important quantity is **entropy** (a measure of unpredictability), defined by $H = -x \ln x - (1 - x) \ln(1 - x)$, for $0 < x < 1$. Find the value of x that maximizes this quantity. Explain why this probability would maximize the measure of unpredictability of errors. (Hint: If $x = 0$ or $x = 1$, are errors unpredictable?)
57. Researchers in a number of fields (including population biology, economics and the study of animal tumors) make use of the Gompertz growth curve, $W(t) = ae^{-be^{-t}}$. As $t \rightarrow \infty$, show that $W(t) \rightarrow a$ and $W'(t) \rightarrow 0$. Find the maximum growth rate.
58. The rate R of an enzymatic reaction as a function of the substrate concentration $[S]$ is given by $R = \frac{[S]R_m}{K_m + [S]}$, where R_m and K_m are constants. K_m is called the Michaelis constant and R_m is referred to as the maximum reaction rate. Show that R_m is not a proper maximum in that the reaction rate can never be equal to R_m .

59. Suppose a painting hangs on a wall as in the figure. The frame extends from 6 ft to 8 ft above the floor. A person whose eyes are 5 ft above the ground stands x feet from the wall and views the painting, with a viewing angle A formed by the ray from the person's eye to the top of the frame and the ray from the person's eye to the bottom of the bottom of the frame. Find the value of x that maximizes the viewing angle A .



What changes if the person's eyes are 6 feet above the ground?

60. Suppose that a hockey player is shooting at a 6-foot-wide net from a distance of d feet away from the goal line and 4 ft to the side of the center line. (a) Find the distance d that maximizes the shooting angle. (b) Repeat part (a) with the shooter 2 ft to the side of the center line. Explain why the answer is so different. (c) Repeat part (a) with the goalie blocking all but the far 2 ft of the goal.



EXPLORATORY EXERCISES

1. Explore the graphs of e^{-x} , xe^{-x} , x^2e^{-x} and x^3e^{-x} . Find all local extrema and use l'Hôpital's Rule to determine the behavior as $x \rightarrow \infty$. You can think of the graph of $x^n e^{-x}$ as showing the results of a tug-of-war: $x^n \rightarrow \infty$ as $x \rightarrow \infty$ but $e^{-x} \rightarrow 0$ as $x \rightarrow \infty$. Describe the graph of $x^n e^{-x}$ in terms of this tug-of-war.
2. Johannes Kepler (1630–1711) is best known as an astronomer, especially for his three laws of planetary motion. However, he was also brilliant mathematically. While serving in Austrian Emperor Matthew I's court, Kepler observed the ability of certain Austrians to quickly and mysteriously compute the capacities of a variety of casks. Each cask (barrel) had a hole in the middle of its side (see Figure a). The Austrian would insert a rod in the hole until

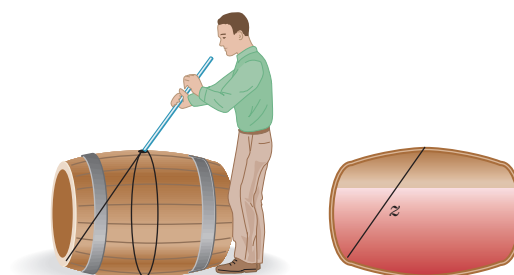


FIGURE a

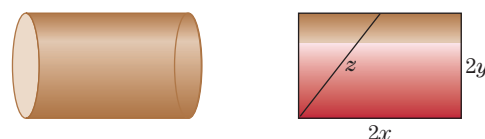


FIGURE b

it hit the far corner and then announce the volume. Kepler first analyzed the problem for a cylindrical barrel (see Figure b). The volume of a cylinder is $V = \pi r^2 h$. In Figure b, $r = y$ and $h = 2x$ so $V = 2\pi y^2 x$. Call the rod measurement z . By the Pythagorean Theorem, $x^2 + (2y)^2 = z^2$. Kepler's mystery was how to compute V given only z . The key observation made by Kepler was that Austrian casks were made with the same height-to-diameter ratio (for us, x/y). Let $t = x/y$ and show that $z^2/y^2 = t^2 + 4$.

Use this to replace y^2 in the volume formula. Then replace x with $\sqrt{z^2 + 4y^2}$. Show that $V = \frac{2\pi z^3 t}{(4 + t^2)^{3/2}}$. In this formula, t is a constant, so the Austrian could measure z and quickly estimate the volume. We haven't told you yet what t equals. Kepler assumed that the Austrians would have made a smart choice for this ratio. Find the value of t that maximizes the volume for a given z . This is, in fact, the ratio used in the construction of Austrian casks!

Increasing and Decreasing Functions

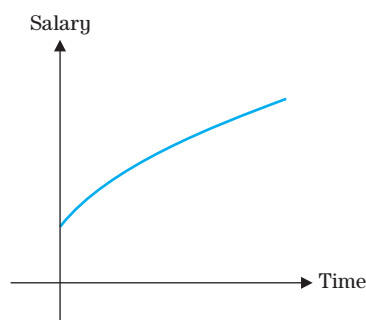


FIGURE 4.40
Increasing salary

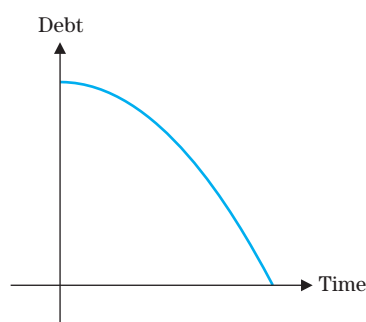


FIGURE 4.41
Decreasing debt

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In section 4.3, we determined that local extrema occur only at critical numbers. However, not all critical numbers correspond to local extrema. In this section, we see how to determine which critical numbers correspond to local extrema. At the same time, we'll learn more about the connection between the derivative and graphing.

We are all familiar with the terms *increasing* and *decreasing*. If your employer informs you that your salary will be increasing steadily over the term of your employment, you have in mind that as time goes on, your salary will rise something like Figure 4.40. If you take out a loan to purchase a car, once you start paying back the loan, your indebtedness will decrease over time. If you plotted your debt against time, the graph might look something like Figure 4.41.

We now carefully define these notions. Notice that Definition 4.1 is merely a formal statement of something you already understand.

DEFINITION 4.1

A function f is **increasing** on an interval I if for every $x_1, x_2 \in I$ with $x_1 < x_2$, $f(x_1) < f(x_2)$ [i.e., $f(x)$ gets larger as x gets larger].

A function f is **decreasing** on the interval I if for every $x_1, x_2 \in I$ with $x_1 < x_2$, $f(x_1) > f(x_2)$ [i.e., $f(x)$ gets smaller as x gets larger].

While anyone can look at a graph of a function and immediately see where that function is increasing and decreasing, the challenge is to determine where a function is increasing and decreasing, given only a mathematical formula for the function. For example, can you determine where $f(x) = x^2 \sin x$ is increasing and decreasing, *without* looking at a graph? Look carefully at Figure 4.42 (on the following page) to see if you can notice what happens at every point at which the function is increasing or decreasing.

Observe that on intervals where the tangent lines have positive slope, f is increasing, while on intervals where the tangent lines have negative slope, f is decreasing. Of course, the slope of the tangent line at a point is given by the value of the derivative at that point. So, whether a function is increasing or decreasing on an interval seems to be determined by the sign of its derivative on that interval. We now state a theorem that makes this connection precise.

THEOREM 4.1

Suppose that f is differentiable on an interval I .

- (i) If $f'(x) > 0$ for all $x \in I$, then f is increasing on I .
- (ii) If $f'(x) < 0$ for all $x \in I$, then f is decreasing on I .

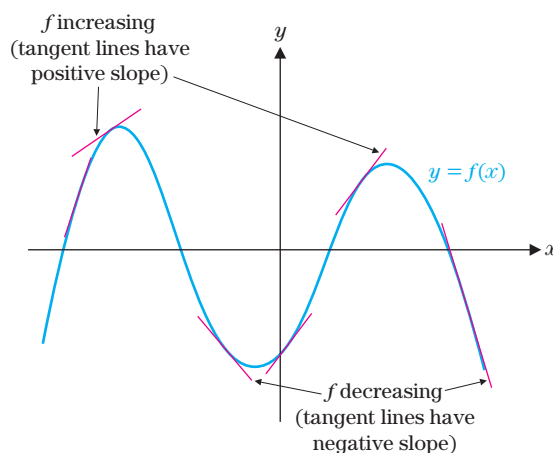


FIGURE 4.42
Increasing and decreasing

PROOF

(i) Pick any two points $x_1, x_2 \in I$, with $x_1 < x_2$. Applying the Mean Value Theorem to f on the interval (x_1, x_2) , we get

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c), \quad (4.1)$$

for some number $c \in (x_1, x_2)$. (Why can we apply the Mean Value Theorem here?) By hypothesis, $f'(c) > 0$ and since $x_1 < x_2$ (so that $x_2 - x_1 > 0$), we have from (4.1) that

$$0 < f(x_2) - f(x_1)$$

or

$$f(x_1) < f(x_2). \quad (4.2)$$

Since (4.2) holds for all $x_1 < x_2$, f is increasing on I .

The proof of (ii) is nearly identical and is left as an exercise. ■

What You See May Not Be What You Get

One aim here and in sections 4.5 and 4.6 is to learn how to draw representative graphs of functions (i.e., graphs that display all of the significant features of a function: where it is increasing or decreasing, any extrema, asymptotes and two features we'll introduce in section 4.5: concavity and inflection points). We draw each graph in a particular viewing *window* (i.e., a particular range of x - and y -values). In the case of computer- or calculator-generated graphs, the window is often chosen by the machine. To uncover when significant features are hidden outside of a given window or to determine the precise locations of features that we can see in a given window, we need some calculus.

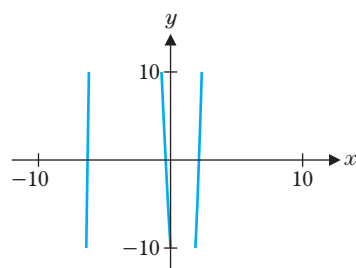


FIGURE 4.43
 $y = 2x^3 + 9x^2 - 24x - 10$

EXAMPLE 4.1 Drawing a Graph

Draw a graph of $f(x) = 2x^3 + 9x^2 - 24x - 10$ showing all local extrema.

Solution Many graphing calculators use the default window defined by $-10 \leq x \leq 10$ and $-10 \leq y \leq 10$. Using this window, the graph of $y = f(x)$ looks like that displayed in Figure 4.43, although the three segments shown are not particularly revealing. Instead of blindly manipulating the window in the hope that a reasonable graph will magically appear, we stop briefly to determine where the function is increasing and decreasing. We have

$$\begin{aligned} f'(x) &= 6x^2 + 18x - 24 = 6(x^2 + 3x - 4) \\ &= 6(x - 1)(x + 4). \end{aligned}$$

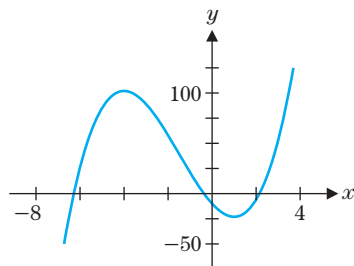
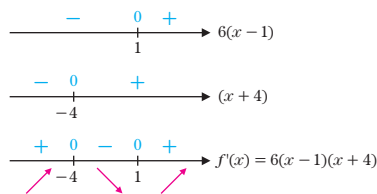


FIGURE 4.44a
 $y = 2x^3 + 9x^2 - 24x - 10$

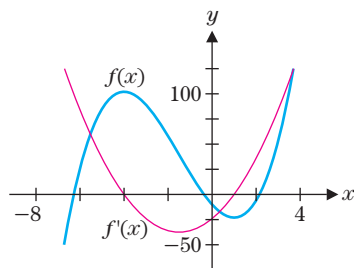


FIGURE 4.44b
 $y = f(x)$ and $y = f'(x)$

Note that the critical numbers (1 and -4) are the only possible locations for local extrema. We can see where the two factors and consequently the derivative are positive and negative from the number lines displayed in the margin. From this, note that

$$f'(x) > 0 \text{ on } (-\infty, -4) \cup (1, \infty) \quad f \text{ increasing}$$

and

$$f'(x) < 0 \text{ on } (-4, 1). \quad f \text{ decreasing.}$$

For convenience, we have placed arrows indicating where the function is increasing and decreasing beneath the last number line. In Figure 4.44a, we redraw the graph in the window defined by $-8 \leq x \leq 4$ and $-50 \leq y \leq 125$. Here, we have selected the y -range so that the critical points $(-4, 102)$ and $(1, -23)$ are displayed. Since f is increasing on all of $(-\infty, -4)$, we know that the function is still increasing to the left of the portion displayed in Figure 4.44a. Likewise, since f is increasing on all of $(1, \infty)$, we know that the function continues to increase to the right of the displayed portion. In Figure 4.44b, we have plotted both $y = f(x)$ (shown in blue) and $y = f'(x)$ (shown in red). Notice the connection between the two graphs. When $f'(x) > 0$, f is increasing; when $f'(x) < 0$, f is decreasing and also notice what happens to $f'(x)$ at the local extrema of f . (We'll say more about this shortly.) ■

You may be tempted to think that you can draw graphs by machine and with a little fiddling with the graphing window, get a reasonable looking graph. Unfortunately, this frequently isn't enough. For instance, while it's clear that the graph in Figure 4.43 is incomplete, the initial graph in example 4.2 has a familiar shape and may look reasonable, but it is incorrect. The calculus tells you what features you should expect to see in a graph. Without it, you're simply taking a shot in the dark.

EXAMPLE 4.2 Uncovering Hidden Behavior in a Graph

Graph $f(x) = 3x^4 + 40x^3 - 0.06x^2 - 1.2x$ showing all local extrema.

Solution The default graph drawn by our computer algebra system is shown in Figure 4.45a, while a common default graphing calculator graph is shown in Figure 4.45b. You can certainly make Figure 4.45b look more like Figure 4.45a by adjusting the window some. But with some calculus, you can discover features that are hidden in both graphs.

First, notice that

$$\begin{aligned} f'(x) &= 12x^3 + 120x^2 - 0.12x - 1.2 \\ &= 12(x^2 - 0.01)(x + 10) \\ &= 12(x - 0.1)(x + 0.1)(x + 10). \end{aligned}$$

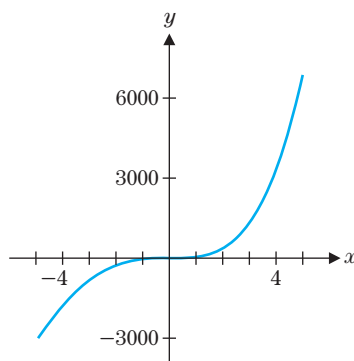
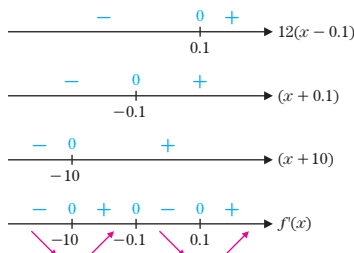


FIGURE 4.45a
Default CAS graph of
 $y = 3x^4 + 40x^3 - 0.06x^2 - 1.2x$

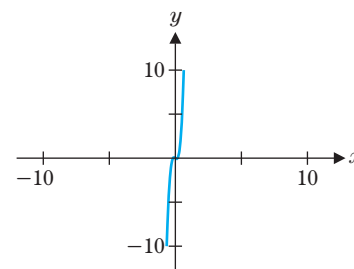


FIGURE 4.45b
Default calculator graph of
 $y = 3x^4 + 40x^3 - 0.06x^2 - 1.2x$

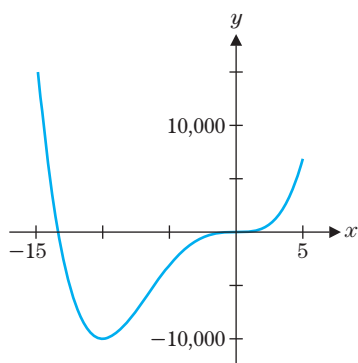


FIGURE 4.46a

The global behavior of
 $f(x) = 3x^4 + 40x^3 - 0.06x^2 - 1.2x$

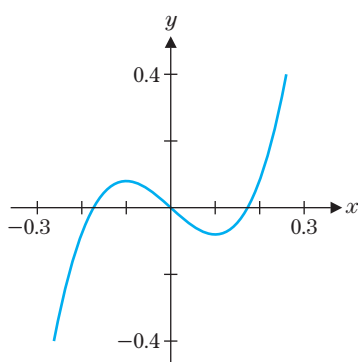


FIGURE 4.46b

Local behavior of
 $f(x) = 3x^4 + 40x^3 - 0.06x^2 - 1.2x$

We show number lines for the three factors in the margin (on the preceding page). Observe that

$$f'(x) \begin{cases} > 0 \text{ on } (-10, -0.1) \cup (0.1, \infty) & f \text{ increasing.} \\ < 0 \text{ on } (-\infty, -10) \cup (-0.1, 0.1). & f \text{ decreasing.} \end{cases}$$

Since both of the graphs in Figures 4.45a and 4.45b suggest that f is increasing for all x , neither of these graphs is adequate. As it turns out, no single graph captures all of the behavior of this function. However, by increasing the range of x -values to the interval $[-15, 5]$, we get the graph seen in Figure 4.46a. This shows what we refer to as the **global** behavior of the function. Here, you can see the local minimum at $x = -10$, which was missing in our earlier graphs, but the behavior for values of x close to zero is not clear. To see this, we need a separate graph, restricted to a smaller range of x -values, as seen in Figure 4.46b. Notice that here, we can clearly see the behavior of the function for x close to zero. In particular, the local maximum at $x = -0.1$ and the local minimum at $x = 0.1$ are clearly visible. We often say that a graph such as Figure 4.46b shows the **local** behavior of the function. In Figures 4.47a and 4.47b, we show graphs indicating the global and local behavior of $f(x)$ (in blue) and $f'(x)$ (in red) on the same set of axes. Pay particular attention to the behavior of $f'(x)$ in the vicinity of local extrema of $f(x)$.

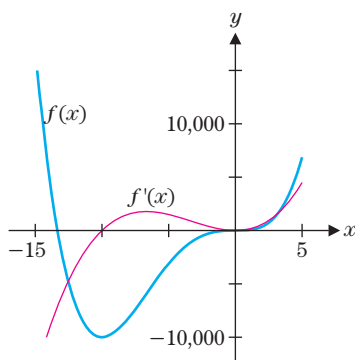


FIGURE 4.47a

$y = f(x)$ and $y = f'(x)$
 (global behavior)

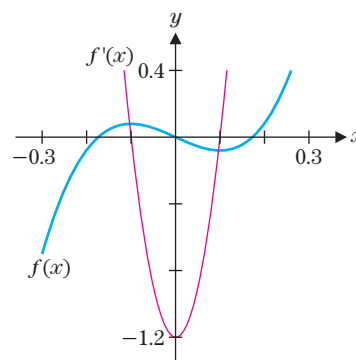


FIGURE 4.47b

$y = f(x)$ and $y = f'(x)$
 (local behavior)

You may have already noticed a connection between local extrema and the intervals on which a function is increasing and decreasing. We state this in Theorem 4.2.

THEOREM 4.2 (First Derivative Test)

Suppose that f is continuous on the interval $[a, b]$ and $c \in (a, b)$ is a critical number.

- (i) If $f'(x) > 0$ for all $x \in (a, c)$ and $f'(x) < 0$ for all $x \in (c, b)$ (i.e., f changes from increasing to decreasing at c), then $f(c)$ is a local maximum.
- (ii) If $f'(x) < 0$ for all $x \in (a, c)$ and $f'(x) > 0$ for all $x \in (c, b)$ (i.e., f changes from decreasing to increasing at c), then $f(c)$ is a local minimum.
- (iii) If $f'(x)$ has the *same* sign on (a, c) and (c, b) , then $f(c)$ is *not* a local extremum.

It's easiest to think of this result graphically. If f is increasing to the left of a critical number and decreasing to the right, then there must be a local maximum at the critical number. (See Figure 4.48a.) Likewise, if f is decreasing to the left of a critical number and increasing to the right, then there must be a local minimum at the critical number. (See Figure 4.48b.) This suggests a proof of the theorem; the job of writing out all of the details is left as an exercise.

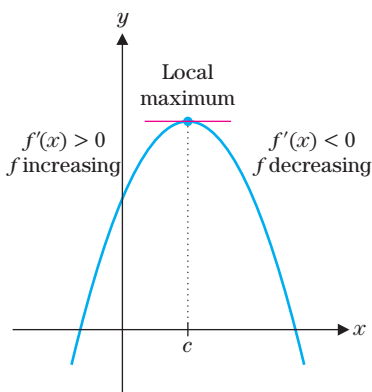


FIGURE 4.48a
Local maximum

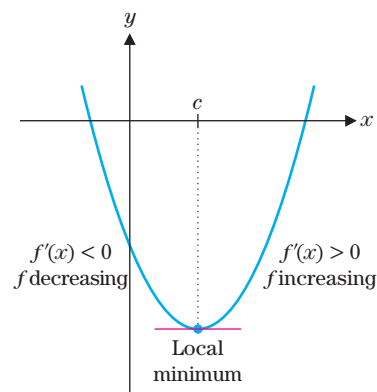


FIGURE 4.48b
Local minimum

EXAMPLE 4.3 Finding Local Extrema Using the First Derivative Test

Find the local extrema of the function from example 4.1, $f(x) = 2x^3 + 9x^2 - 24x - 10$.

Solution We had found in example 4.1 that

$$f'(x) \begin{cases} > 0, \text{ on } (-\infty, -4) \cup (1, \infty) & f \text{ increasing.} \\ < 0, \text{ on } (-4, 1). & f \text{ decreasing.} \end{cases}$$

It now follows from the First Derivative Test that f has a local maximum located at $x = -4$ and a local minimum located at $x = 1$.

Theorem 4.2 works equally well for a function with critical points where the derivative is undefined.

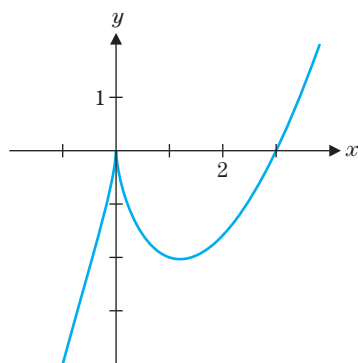
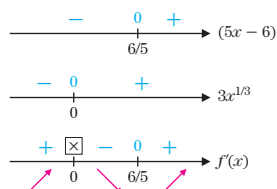


FIGURE 4.49
 $y = x^{5/3} - 3x^{2/3}$

EXAMPLE 4.4 Finding Local Extrema of a Function with Fractional Exponents

Find the local extrema of $f(x) = x^{5/3} - 3x^{2/3}$.

Solution We have

$$f'(x) = \frac{5}{3}x^{2/3} - 3\left(\frac{2}{3}\right)x^{-1/3} = \frac{5x - 6}{3x^{1/3}},$$

so that the critical numbers are $\frac{6}{5}$ [$f'(\frac{6}{5}) = 0$] and 0 [$f'(0)$ is undefined]. Again drawing number lines for the factors, we determine where f is increasing and decreasing. Here, we have placed an $\boxtimes$ above the 0 on the number line for $f'(x)$ to indicate that f' is not defined at $x = 0$. From this, we can see at a glance where f' is positive and negative:

$$f'(x) \begin{cases} > 0, \text{ on } (-\infty, 0) \cup (\frac{6}{5}, \infty) & f \text{ increasing.} \\ < 0, \text{ on } (0, \frac{6}{5}). & f \text{ decreasing.} \end{cases}$$

Consequently, f has a local maximum at $x = 0$ and a local minimum at $x = \frac{6}{5}$. These local extrema are both clearly visible in the graph in Figure 4.49.

EXAMPLE 4.5 Finding Local Extrema Approximately

Find the local extrema of $f(x) = x^4 + 4x^3 - 5x^2 - 31x + 29$ and draw a graph.

Solution A graph of $y = f(x)$ using the most common graphing calculator default window appears in Figure 4.50 (on the following page). Without further analysis, we do not know whether this graph shows all of the significant behavior of the

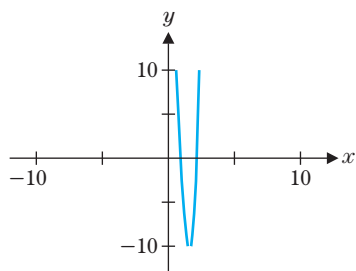


FIGURE 4.50

$$f(x) = x^4 + 4x^3 - 5x^2 - 31x + 29$$

function. [Note that some fourth-degree polynomials (e.g., $f(x) = x^4$) have graphs that look very much like the one in Figure 4.50.] First, we compute

$$f'(x) = 4x^3 + 12x^2 - 10x - 31.$$

However, this derivative does not easily factor. A graph of $y = f'(x)$ (see Figure 4.51) reveals three zeros, one near each of $x = -3$, -1.5 and 1.5 . Since a cubic polynomial has at most three zeros, there are no others. Using Newton's method or some other rootfinding method [applied to $f'(x)$], we can find approximations to the three zeros of f' . We get $a \approx -2.96008$, $b \approx -1.63816$ and $c \approx 1.59824$. From Figure 4.51, we can see that

$$f'(x) > 0 \text{ on } (a, b) \cup (c, \infty) \quad f \text{ increasing.}$$

and

$$f'(x) < 0 \text{ on } (-\infty, a) \cup (b, c). \quad f \text{ decreasing.}$$

From the First Derivative Test, there is a local minimum at $a \approx -2.96008$, a local maximum at $b \approx -1.63816$ and a local minimum at $c \approx 1.59824$. Since only the local minimum at $x = c$ is visible in the graph in Figure 4.50, this graph is inadequate. By narrowing the range of displayed x -values and widening the range of displayed y -values, we obtain the far more useful graph seen in Figure 4.52. Note that the local minimum at $x = c \approx 1.59824$ is also the *absolute* minimum.

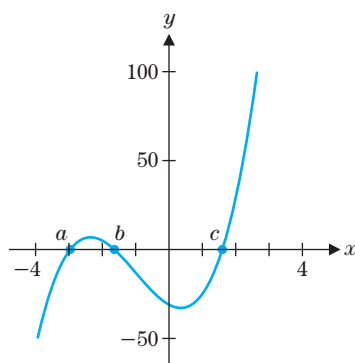


FIGURE 4.51

$$f'(x) = 4x^3 + 12x^2 - 10x - 31$$

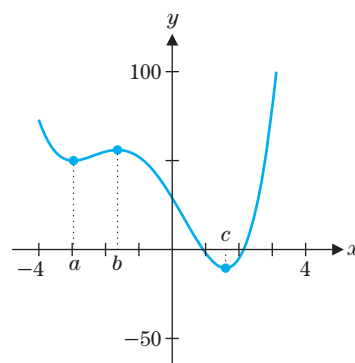


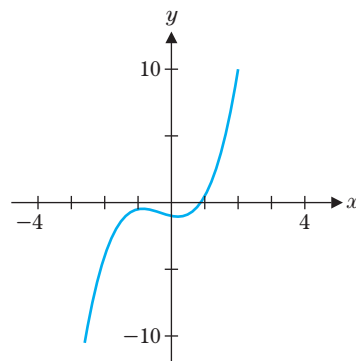
FIGURE 4.52

$$f(x) = x^4 + 4x^3 - 5x^2 - 31x + 29$$

EXERCISES 4.4

WRITING EXERCISES

- Suppose that $f(0) = 2$ and f is an increasing function. To sketch the graph of $y = f(x)$, you could start by plotting the point $(0, 2)$. Filling in the graph to the left, would you move your pencil up or down? How does this fit with the definition of increasing?
- Suppose you travel east on an east-west interstate highway. You reach your destination, stay a while and then return home. Explain the First Derivative Test in terms of your velocities (positive and negative) on this trip.
- Suppose that you have a differentiable function f with two distinct critical numbers. Your computer has shown you a graph that looks like the one in the figure.



Discuss the possibility that this is a representative graph: that is, is it possible that there are any important points not shown in this window?

4. Suppose that the function in exercise 3 has three distinct critical numbers. Explain why the graph is not a representative graph. Discuss how you would change the graphing window to show the rest of the graph.

In exercises 1–10, find (by hand) the intervals where the function is increasing and decreasing. Use this information to determine all local extrema and sketch a graph.

- | | |
|--------------------------|------------------------------|
| 1. $y = x^3 - 3x + 2$ | 2. $y = x^3 + 2x^2 + 1$ |
| 3. $y = x^4 - 8x^2 + 1$ | 4. $y = x^3 - 3x^2 - 9x + 1$ |
| 5. $y = (x + 1)^{2/3}$ | 6. $y = (x - 1)^{1/3}$ |
| 7. $y = \sin x + \cos x$ | 8. $y = \sin^2 x$ |
| 9. $y = e^{x^2-1}$ | 10. $y = \ln(x^2 - 1)$ |

In exercises 11–20, find (by hand) all critical numbers and use the First Derivative Test to classify each as the location of a local maximum, local minimum or neither.

- | | |
|-----------------------------|---|
| 11. $y = x^4 + 4x^3 - 2$ | 12. $y = x^5 - 5x^2 + 1$ |
| 13. $y = xe^{-2x}$ | 14. $y = x^2 e^{-x}$ |
| 15. $y = \tan^{-1}(x^2)$ | 16. $y = \sin^{-1}\left(1 - \frac{1}{x^2}\right)$ |
| 17. $y = \frac{x}{1 + x^3}$ | 18. $y = \frac{x}{1 + x^4}$ |
| 19. $y = \sqrt{x^3 + 3x^2}$ | 20. $y = x^{4/3} + 4x^{1/3}$ |

In exercises 21–26, approximate the x -coordinates of all extrema and sketch graphs showing global and local behavior of the function.

21. $y = x^4 - 15x^3 - 2x^2 + 40x - 2$
 22. $y = x^4 - 16x^3 - 0.1x^2 + 0.5x - 1$
 23. $y = x^5 - 200x^3 + 605x - 2$
 24. $y = x^4 - 0.5x^3 - 0.02x^2 + 0.02x + 1$
 25. $y = (x^2 + x + 0.45)e^{-2x}$
 26. $y = x^5 \ln 8x^2$

In exercises 27–32, sketch a graph of a function with the given properties.

27. $f(0) = 1$, $f(2) = 5$, $f'(x) < 0$ for $x < 0$ and $x > 2$, $f'(x) > 0$ for $0 < x < 2$.
 28. $f(-1) = 1$, $f(2) = 5$, $f'(x) < 0$ for $x < -1$ and $x > 2$, $f'(x) > 0$ for $-1 < x < 2$, $f'(-1) = 0$, $f'(2)$ does not exist.
 29. $f(3) = 0$, $f'(x) < 0$ for $x < 0$ and $x > 3$, $f'(x) > 0$ for $0 < x < 3$, $f'(3) = 0$, $f(0)$ and $f'(0)$ do not exist.

30. $f(1) = 0$, $\lim_{x \rightarrow \infty} f(x) = 2$, $f'(x) < 0$ for $x < 1$, $f'(x) > 0$ for $x > 1$, $f'(1) = 0$.

31. $f(-1) = f(2) = 0$, $f'(x) < 0$ for $x < -1$ and $0 < x < 2$ and $x > 2$, $f'(x) > 0$ for $-1 < x < 0$, $f'(-1)$ does not exist, $f'(2) = 0$.

32. $f(0) = 0$, $f(3) = -1$, $f'(x) < 0$ for $x > 3$, $f'(x) > 0$ for $x < 0$ and $0 < x < 1$ and $1 < x < 3$, $f'(0) = 0$, $f'(1)$ does not exist and $f'(3) = 0$.

In exercises 33–38, find (by hand) all asymptotes and extrema, and sketch a graph.

- | | |
|------------------------------------|-------------------------------------|
| 33. $y = \frac{x}{x^2 - 1}$ | 34. $y = \frac{x^2}{x^2 - 1}$ |
| 35. $y = \frac{x^2}{x^2 - 4x + 3}$ | 36. $y = \frac{x}{1 - x^4}$ |
| 37. $y = \frac{x}{\sqrt{x^2 + 1}}$ | 38. $y = \frac{x^2 + 2}{(x + 1)^2}$ |

In exercises 39–42, estimate critical numbers and sketch graphs showing both global and local behavior.

- | | |
|----------------------------------|-----------------------------------|
| 39. $y = \frac{x - 30}{x^4 - 1}$ | 40. $y = \frac{x^2 - 8}{x^4 - 1}$ |
| 41. $y = \frac{x + 60}{x^2 + 1}$ | 42. $y = \frac{x - 60}{x^2 - 1}$ |

43. Give a graphical example showing that the following statement is false. If $f(0) = 4$ and f is a decreasing function, then the equation $f(x) = 0$ has exactly one solution.

44. Assume that f is an increasing function with inverse function f^{-1} . Show that f^{-1} is also an increasing function.

45. State the domain for $\sin^{-1} x$ and determine where it is increasing and decreasing.

46. State the domain for $\sin^{-1}\left(\frac{2}{\pi} \tan^{-1} x\right)$ and determine where it is increasing and decreasing.

47. If f and g are both increasing functions, is it true that $f(g(x))$ is also increasing? Either prove that it is true or give an example that proves it false.

48. If f and g are both increasing functions with $f(5) = 0$, find the maximum and minimum of the following values: $g(1)$, $g(4)$, $g(f(1))$, $g(f(4))$.

49. For $f(x) = \begin{cases} x + 2x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ show that $f'(0) > 0$,

but that f is not increasing in any interval around 0. Explain why this does not contradict Theorem 4.1.

50. For $f(x) = x^3$, show that f is increasing in any interval around 0, but $f'(0) = 0$. Explain why this does not contradict Theorem 4.1.

51. Prove Theorem 4.2 (the First Derivative Test).

52. Give a graphical argument that if $f(a) = g(a)$ and $f'(x) > g'(x)$ for all $x > a$, then $f(x) > g(x)$ for all $x > a$. Use the Mean Value Theorem to prove it.

In exercises 53–56, use the result of exercise 52 to verify the inequality.

53. $2\sqrt{x} > 3 - \frac{1}{x}$ for $x > 1$ 54. $x > \sin x$ for $x > 0$
 55. $e^x > x + 1$ for $x > 0$ 56. $x - 1 > \ln x$ for $x > 1$

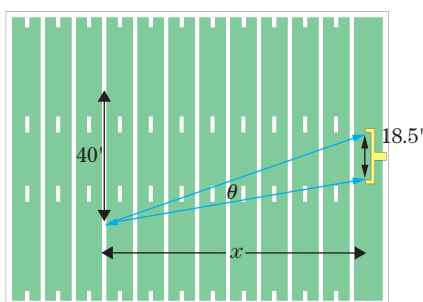
57. Show that $f(x) = x^3 + bx^2 + cx + d$ is an increasing function if $b^2 \leq 3c$. Find a condition on the coefficients b and c that guarantees that $f(x) = x^5 + bx^3 + cx + d$ is an increasing function.
 58. Suppose that f and g are differentiable functions and $x = c$ is a critical number of both functions. Either prove (if it is true) or disprove (with a counterexample) that the composition $f \circ g$ also has a critical number at $x = c$.

APPLICATIONS

59. Suppose that the total sales of a product after t months is given by $s(t) = \sqrt{t+4}$ thousand dirhams. Compute and interpret $s'(t)$.
 60. In exercise 59, show that $s'(t) > 0$ for all $t > 0$. Explain in business terms why it is impossible to have $s'(t) < 0$.
 61. The table shows the coefficient of friction μ of ice as a function of temperature. The lower μ is, the more “slippery” the ice is. Estimate $\mu'(C)$ at (a) $C = -10$ and (b) $C = -6$. If skating warms the ice, does it get easier or harder to skate? Briefly explain.

$^{\circ}\text{C}$	-12	-10	-8	-6	-4	-2
μ	0.0048	0.0045	0.0043	0.0045	0.0048	0.0055

62. For a college football field with the dimensions shown, the angle θ for kicking a field goal from a (horizontal) distance



of x feet from the goal post is given by $\theta(x) = \tan^{-1}(29.25/x) - \tan^{-1}(10.75/x)$. Show that $f(t) = \frac{t}{a^2 + t^2}$ is increasing for $a > t$ and use this fact to show that $\theta(x)$ is a decreasing function for $x \geq 30$. Announcers often say that for a short field goal ($50 \leq x \leq 60$), a team can improve the angle by backing up 5 yards with a penalty. Is this true?

EXPLORATORY EXERCISES

- In this exercise, we look at the ability of fireflies to synchronize their flashes. Let the function f represent an individual firefly's rhythm, so that the firefly flashes whenever $f(t)$ equals an integer. Let $e(t)$ represent the rhythm of a neighboring firefly, where again $e(t) = n$, for some integer n , whenever the neighbor flashes. One model of the interaction between fireflies is $f'(t) = \omega + A \sin[e(t) - f(t)]$ for constants ω and A . If the fireflies are synchronized [$e(t) = f(t)$], then $f'(t) = \omega$, so the fireflies flash every $1/\omega$ time units. Assume that the difference between $e(t)$ and $f(t)$ is less than π . Show that if $f(t) < e(t)$, then $f'(t) > \omega$. Explain why this means that the individual firefly is speeding up its flash to match its neighbor. Similarly, discuss what happens if $f(t) > e(t)$.
- In a sport like football or hockey where ties are possible, the probability that the stronger team wins depends in an interesting way on the number of goals scored. Suppose that at any point, the probability that team A scores the next goal is p , where $0 < p < 1$. If 2 goals are scored, a 1-1 tie could result from team A scoring first (probability p) and then team B tying the score (probability $1 - p$), or vice versa. The probability of a tie in a 2-goal game is then $2p(1 - p)$. Similarly, the probability of a 2-2 tie in a 4-goal game is $\frac{4 \cdot 3}{2 \cdot 1} p^2(1 - p)^2$, the probability of a 3-3 tie in a 6-goal game is $\frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} p^3(1 - p)^3$ and so on. As the number of goals increases, does the probability of a tie increase or decrease? To find out, first show that $\frac{(2x+2)(2x+1)}{(x+1)^2} < 4$ for $x > 0$ and $x(1 - x) \leq \frac{1}{4}$ for $0 \leq x \leq 1$. Use these inequalities to show that the probability of a tie decreases as the (even) number of goals increases. In a 1-goal game, the probability that team A wins is p . In a 2-goal game, the probability that team A wins is p^2 . In a 3-goal game, the probability that team A wins is $p^3 + 3p^2(1 - p)$. In a 4-goal game, the probability that team A wins is $p^4 + 4p^3(1 - p)$. In a 5-goal game, the probability that team A wins is $p^5 + 5p^4(1 - p) + \frac{5 \cdot 4}{2 \cdot 1} p^3(1 - p)^2$. Explore the extent to which the probability that team A wins increases as the number of goals increases.

Concavity and the Second Derivative Test

In the previous section, we saw how to determine where a function is increasing and decreasing and how this relates to drawing a graph of the function. Unfortunately, simply knowing where a function increases and decreases is not sufficient to draw a good graph. In Figures 4.53a and 4.53b, we show two very different shapes of increasing functions joining the same two points.

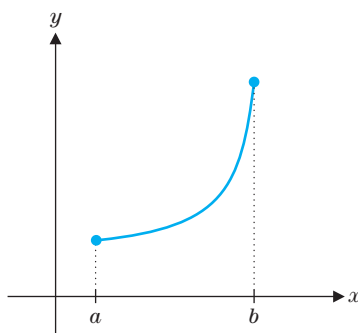


FIGURE 4.53a
Increasing function

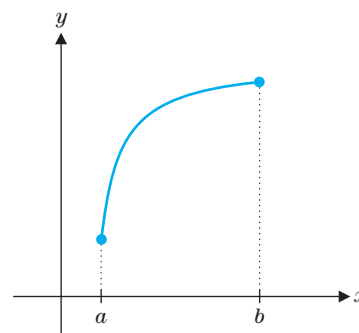


FIGURE 4.53b
Increasing function

Note that the rate of growth in Figure 4.53a, is increasing, while the rate of growth depicted in Figure 4.53b is decreasing. As a further illustration of this, Figures 4.54a and 4.54b are the same as Figures 4.53a and 4.53b, respectively, but with a few tangent lines drawn in.

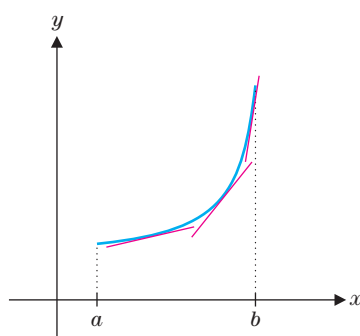


FIGURE 4.54a
Concave up, increasing

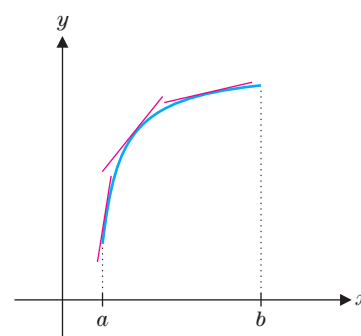


FIGURE 4.54b
Concave down, increasing

Although all of the tangent lines have positive slope [since $f'(x) > 0$], the slopes of the tangent lines in Figure 4.54a are increasing, while those in Figure 4.54b are decreasing. We call the graph in Figure 4.54a **concave up** and the graph in Figure 4.54b **concave down**. The situation is similar for decreasing functions. In Figures 4.55a and 4.55b, we

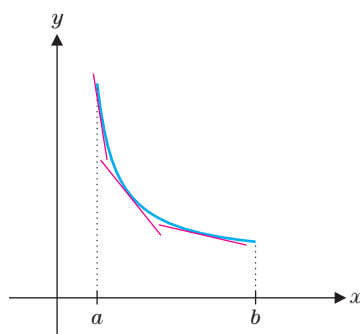


FIGURE 4.55a
Concave up, decreasing

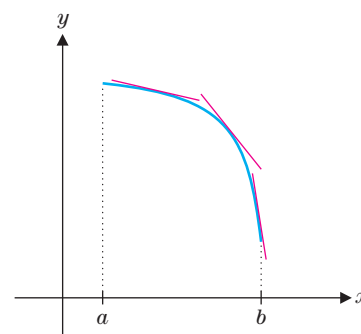


FIGURE 4.55b
Concave down, decreasing

show two different shapes of decreasing functions. The one shown in Figure 4.55a is concave up (slopes of tangent lines increasing) and the one shown in Figure 4.55b is concave down (slopes of tangent lines decreasing). We summarize this in Definition 5.1.

DEFINITION 5.1

For a function f that is differentiable on an interval I , the graph of f is

- (i) **concave up** on I if f' is increasing on I .
- (ii) **concave down** on I if f' is decreasing on I .

Note that you can tell when f' is increasing or decreasing from the derivative of f' (i.e., f''). Theorem 5.1 connects concavity with what we already know about increasing and decreasing functions. The proof is a straightforward application of Theorem 4.1 to Definition 5.1.

THEOREM 5.1

Suppose that f'' exists on an interval I .

- (i) If $f''(x) > 0$ on I , then the graph of f is concave up on I .
- (ii) If $f''(x) < 0$ on I , then the graph of f is concave down on I .

EXAMPLE 5.1 Determining Concavity

Determine where the graph of $f(x) = 2x^3 + 9x^2 - 24x - 10$ is concave up and concave down, and draw a graph showing all significant features of the function.

Solution Here, we have $f'(x) = 6x^2 + 18x - 24$

and from our work in example 4.3, we have

$$f'(x) \begin{cases} > 0 \text{ on } (-\infty, -4) \cup (1, \infty) & f \text{ increasing.} \\ < 0 \text{ on } (-4, 1). & f \text{ decreasing.} \end{cases}$$

Further, we have $f''(x) = 12x + 18 \begin{cases} > 0, \text{ for } x > -\frac{3}{2} & \text{Concave up.} \\ < 0, \text{ for } x < -\frac{3}{2}. & \text{Concave down.} \end{cases}$

Using all of this information, we are able to draw the graph shown in Figure 4.56. Notice that at the point $\left(-\frac{3}{2}, f\left(-\frac{3}{2}\right)\right)$, the graph changes from concave down to concave up. Such points are called *inflection points*, which we define more precisely in Definition 5.2. ■

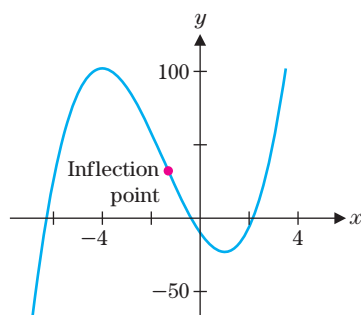


FIGURE 4.56
 $y = 2x^3 + 9x^2 - 24x - 10$

NOTES

If $(c, f(c))$ is an inflection point, then either $f''(c) = 0$ or $f''(c)$ is undefined. So, finding all points where $f''(x)$ is zero or is undefined gives you all possible candidates for inflection points. But beware: not all points where $f''(x)$ is zero or undefined correspond to inflection points.

DEFINITION 5.2

Suppose that f is continuous on the interval (a, b) and that the graph changes concavity at a point $c \in (a, b)$ (i.e., the graph is concave down on one side of c and concave up on the other). Then, the point $(c, f(c))$ is called an **inflection point** of f .

EXAMPLE 5.2 Determining Concavity and Locating Inflection Points

Determine where the graph of $f(x) = x^4 - 6x^2 + 1$ is concave up and concave down, find any inflection points and draw a graph showing all significant features.

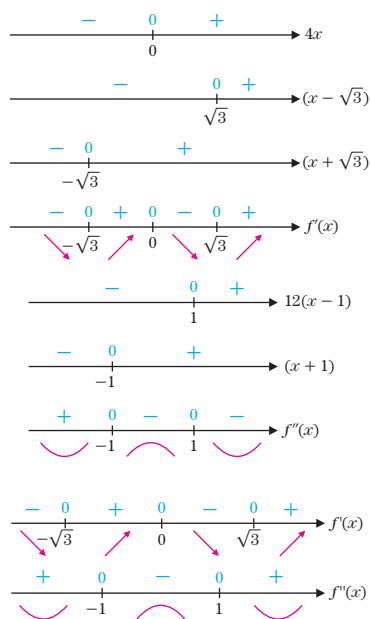


FIGURE 4.57
 $y = x^4 - 6x^2 + 1$

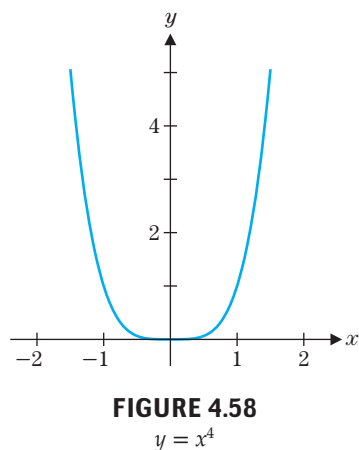


FIGURE 4.58
 $y = x^4$

Solution Here, we have

$$\begin{aligned} f'(x) &= 4x^3 - 12x = 4x(x^2 - 3) \\ &= 4x(x - \sqrt{3})(x + \sqrt{3}). \end{aligned}$$

We have drawn number lines for the factors of $f'(x)$ in the margin. From this, we can see that

$$f'(x) \begin{cases} > 0, \text{ on } (-\sqrt{3}, 0) \cup (\sqrt{3}, \infty) & f \text{ increasing.} \\ < 0, \text{ on } (-\infty, -\sqrt{3}) \cup (0, \sqrt{3}). & f \text{ decreasing.} \end{cases}$$

Next, we have

$$f''(x) = 12x^2 - 12 = 12(x - 1)(x + 1).$$

We have drawn number lines for the two factors in the margin. From this, we can see that

$$f''(x) \begin{cases} > 0, \text{ on } (-\infty, -1) \cup (1, \infty) & \text{Concave up.} \\ < 0, \text{ on } (-1, 1). & \text{Concave down.} \end{cases}$$

For convenience, we have indicated the concavity below the bottom number line, with small concave up and concave down segments. Finally, observe that since the graph changes concavity at $x = -1$ and $x = 1$, there are inflection points located at $(-1, -4)$ and $(1, -4)$. Using all of this information, we are able to draw the graph shown in Figure 4.57. For your convenience, we have reproduced the number lines for $f'(x)$ and $f''(x)$ together above the figure. ■

As we see in example 5.3, having $f''(x) = 0$ does not imply the existence of an inflection point.

EXAMPLE 5.3 A Graph with No Inflection Points

Determine the concavity of $f(x) = x^4$ and locate any inflection points.

Solution There's nothing tricky about this function. We have $f'(x) = 4x^3$ and $f''(x) = 12x^2$. Since $f'(x) > 0$ for $x > 0$ and $f'(x) < 0$ for $x < 0$, we know that f is increasing for $x > 0$ and decreasing for $x < 0$. Further, $f''(x) > 0$ for all $x \neq 0$, while $f''(0) = 0$. So, the graph is concave up for $x \neq 0$. Further, even though $f''(0) = 0$, there is *no* inflection point at $x = 0$. We show a graph of the function in Figure 4.58. ■

We now explore a connection between second derivatives and extrema. Suppose that $f'(c) = 0$ and that the graph of f is concave down in some open interval containing c . Then, near $x = c$, the graph looks like that in Figure 4.59a and hence, $f(c)$ is a local maximum. Likewise, if $f'(c) = 0$ and the graph of f is concave up in some open interval containing c , then near $x = c$, the graph looks like that in Figure 4.59b and hence, $f(c)$ is a local minimum.

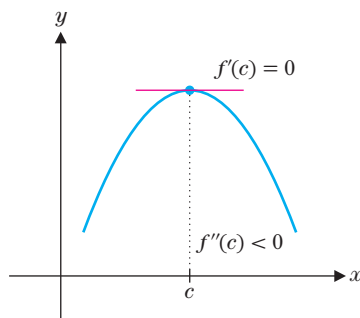


FIGURE 4.59a
Local maximum

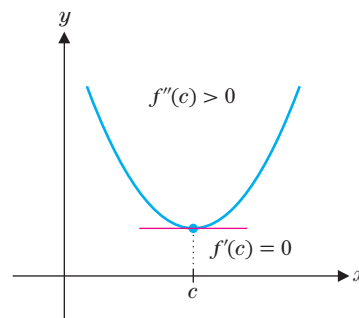


FIGURE 4.59b
Local minimum

We state this more precisely in Theorem 5.2.

THEOREM 5.2 (Second Derivative Test)

Suppose that f'' is continuous on the interval (a, b) and $f'(c) = 0$, for some number $c \in (a, b)$.

- (i) If $f''(c) < 0$, then $f(c)$ is a local maximum.
- (ii) If $f''(c) > 0$, then $f(c)$ is a local minimum.

We leave a formal proof of this theorem as an exercise. When applying the theorem, simply think about Figures 4.59a and 4.59b.

EXAMPLE 5.4 Using the Second Derivative Test to Find Extrema

Use the Second Derivative Test to find the local extrema of $f(x) = x^4 - 8x^2 + 10$.

Solution Here,

$$\begin{aligned} f'(x) &= 4x^3 - 16x = 4x(x^2 - 4) \\ &= 4x(x - 2)(x + 2). \end{aligned}$$

Thus, the critical numbers are $x = 0, 2$ and -2 . We also have

$$f''(x) = 12x^2 - 16$$

and so,

$$\begin{aligned} f''(0) &= -16 < 0, \\ f''(-2) &= 32 > 0 \end{aligned}$$

and

$$f''(2) = 32 > 0.$$

So, by the Second Derivative Test, $f(0)$ is a local maximum and $f(-2)$ and $f(2)$ are local minima. We show a graph of $y = f(x)$ in Figure 4.60. ■

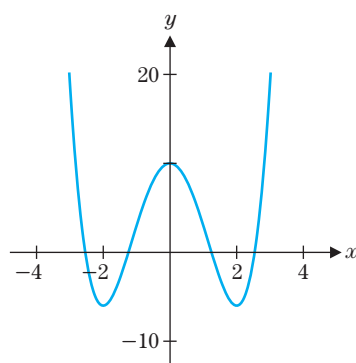


FIGURE 4.60
 $y = x^4 - 8x^2 + 10$

REMARK 5.1

If $f''(c) = 0$ or $f''(c)$ is undefined, the Second Derivative Test yields no conclusion. That is, $f(c)$ may be a local maximum, a local minimum or neither. In this event, we must rely on other methods (such as the First Derivative Test) to determine whether $f(c)$ is a local extremum. We illustrate this with example 5.5.

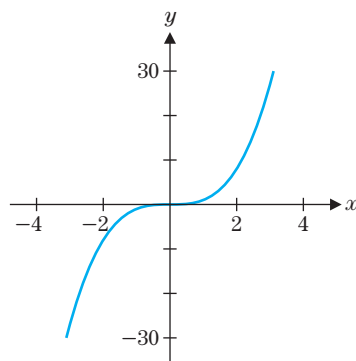


FIGURE 4.61a
 $y = x^3$

EXAMPLE 5.5 Functions for Which the Second Derivative Test Is Inconclusive

Use the Second Derivative Test to try to classify any local extrema for (a) $f(x) = x^3$, (b) $g(x) = (x + 1)^4$ and (c) $h(x) = -x^4$.

Solution (a) Note that $f'(x) = 3x^2$ and $f''(x) = 6x$. So, the only critical number is $x = 0$ and $f''(0) = 0$, also. We leave it as an exercise to show that the point $(0, 0)$ is not a local extremum. (See Figure 4.61a.)

(b) We have $g'(x) = 4(x + 1)^3$ and $g''(x) = 12(x + 1)^2$. Here, the only critical number is $x = -1$ and $g''(-1) = 0$. In this case, though, $g'(x) < 0$ for $x < -1$ and $g'(x) > 0$ for $x > -1$. So, by the First Derivative Test, $(-1, 0)$ is a local minimum. (See Figure 4.61b.)

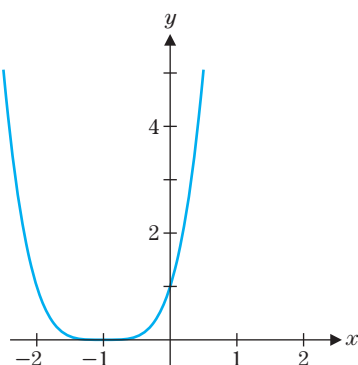


FIGURE 4.61b
 $y = (x + 1)^4$

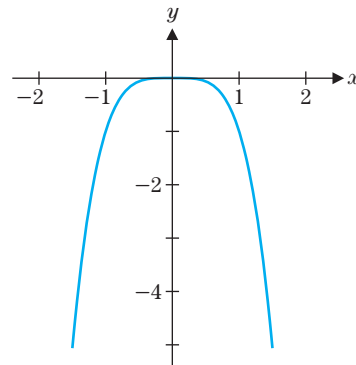


FIGURE 4.61c
 $y = -x^4$

(c) Finally, we have $h'(x) = -4x^3$ and $h''(x) = -12x^2$. Once again, the only critical number is $x = 0$, $h''(0) = 0$ and we leave it as an exercise to show that $(0, 0)$ is a local maximum for h . (See Figure 4.61c.)

We can use first and second derivative information to help produce a meaningful graph of a function, as in example 5.6.

EXAMPLE 5.6 Drawing a Graph of a Rational Function

Draw a graph of $f(x) = x + \frac{25}{x}$, showing all significant features.

Solution The domain of f consists of all real numbers other than $x = 0$. Then,

$$\begin{aligned} f'(x) &= 1 - \frac{25}{x^2} = \frac{x^2 - 25}{x^2} && \text{Add the fractions.} \\ &= \frac{(x - 5)(x + 5)}{x^2}. \end{aligned}$$

So, the only two critical numbers are $x = -5$ and $x = 5$. (Why is $x = 0$ *not* a critical number?)

Looking at the three factors in $f'(x)$, we get the number lines shown in the margin. Thus,

$$f'(x) \begin{cases} > 0, & \text{on } (-\infty, -5) \cup (5, \infty) & f \text{ increasing.} \\ < 0, & \text{on } (-5, 0) \cup (0, 5). & f \text{ decreasing.} \end{cases}$$

Further,

$$f''(x) = \frac{50}{x^3} \begin{cases} > 0, & \text{on } (0, \infty) & \text{Concave up.} \\ < 0, & \text{on } (-\infty, 0). & \text{Concave down.} \end{cases}$$

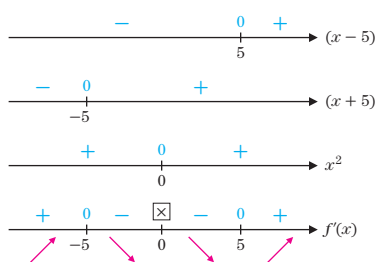
Be careful here. There is *no* inflection point on the graph, even though the graph is concave up on one side of $x = 0$ and concave down on the other. (Why not?) We can now use either the First Derivative Test or the Second Derivative Test to determine the local extrema. Since

$$f''(5) = \frac{50}{125} > 0$$

and

$$f''(-5) = -\frac{50}{125} < 0,$$

there is a local minimum at $x = 5$ and a local maximum at $x = -5$ by the Second Derivative Test. Finally, before we can draw a reasonable graph, we need to know



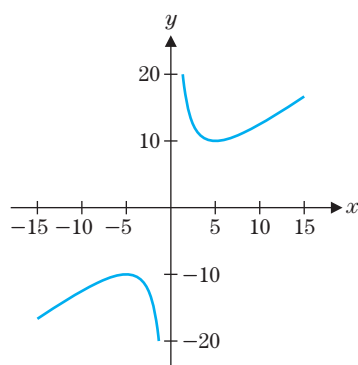


FIGURE 4.62

$$y = x + \frac{25}{x}$$

what happens to the graph near $x = 0$, since 0 is not in the domain of f . We have

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \left(x + \frac{25}{x} \right) = \infty$$

and

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left(x + \frac{25}{x} \right) = -\infty,$$

so that there is a vertical asymptote at $x = 0$. Putting together all of this information, we get the graph shown in Figure 4.62. ■

In example 5.6, we computed $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$ to uncover the behavior of the function near $x = 0$, since $x = 0$ was not in the domain of f . In example 5.7, we'll see that since $x = -2$ is not in the domain of f' (although it is in the domain of f), we must compute $\lim_{x \rightarrow -2^+} f'(x)$ and $\lim_{x \rightarrow -2^-} f'(x)$ to uncover the behavior of the tangent lines near $x = -2$.

EXAMPLE 5.7 A Function with a Vertical Tangent Line at an Inflection Point

Draw a graph of $f(x) = (x + 2)^{1/5} + 4$, showing all significant features.

Solution First, notice that the domain of f is the entire real line. We also have

$$f'(x) = \frac{1}{5}(x + 2)^{-4/5} > 0, \text{ for } x \neq -2.$$

So, f is increasing everywhere, except at $x = -2$ [the only critical number, where $f'(-2)$ is undefined]. This also says that f has no local extrema. Further,

$$f''(x) = -\frac{4}{25}(x + 2)^{-9/5} \begin{cases} > 0, \text{ on } (-\infty, -2) & \text{Concave up.} \\ < 0, \text{ on } (-2, \infty). & \text{Concave down.} \end{cases}$$

So, there is an inflection point at $x = -2$. In this case, $f'(x)$ is undefined at $x = -2$. Since -2 is in the domain of f , but not in the domain of f' , we consider

$$\lim_{x \rightarrow -2^-} f'(x) = \lim_{x \rightarrow -2^-} \frac{1}{5}(x + 2)^{-4/5} = \infty$$

and

$$\lim_{x \rightarrow -2^+} f'(x) = \lim_{x \rightarrow -2^+} \frac{1}{5}(x + 2)^{-4/5} = \infty.$$

This says that the graph has a vertical tangent line at $x = -2$. Putting all of this information together, we get the graph shown in Figure 4.63. ■

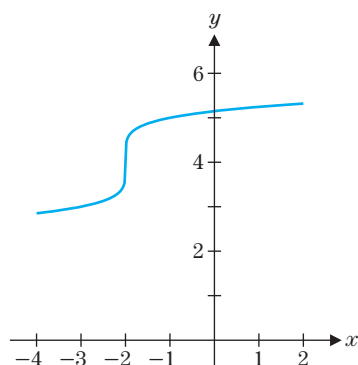


FIGURE 4.63

$$y = (x + 2)^{1/5} + 4$$

EXERCISES 4.5

WRITING EXERCISES

1. It is often said that a graph is concave up if it "holds water." This is certainly true for parabolas like $y = x^2$, but is it true for graphs like $y = 1/x^2$? It can be helpful to put a concept into everyday language, but the danger is in oversimplification. Do you think that "holds water" is helpful? Give your own description of concave up, using everyday language. (Hint: One popular image involves smiles and frowns.)
2. Look up the census population of the United States since 1800. From 1800 to 1900, the numerical increase by decade increased. Argue that this means that the population curve is concave up. From 1960 to 1990, the numerical increase by decade has been approximately constant. Argue that this

means that the population curve is near a point where the curve is neither concave up nor concave down. Why does this not necessarily mean that we are at an inflection point?

3. The goal of investing in the stock market is to buy low and sell high. But, how can you tell whether a price has peaked? Once a stock price goes down, you can see that it *was* at a peak, but then it's too late! Concavity can help. Suppose a stock price is increasing and the price curve is concave up. Why would you suspect that it will continue to rise? Is this a good time to buy? Now, suppose the price is increasing but the curve is concave down. Why should you be preparing to sell? Finally, suppose the price is decreasing. If the curve is concave up, should you buy or sell? What if the curve is concave down?

4. Suppose that $f(t)$ is the amount of money in your bank account at time t . Explain in terms of spending and saving what would cause $f(t)$ to be decreasing and concave down; increasing and concave up; decreasing and concave up.

In exercises 1–8, determine the intervals where the graph of the given function is concave up and concave down, and identify inflection points.

1. $f(x) = x^3 - 3x^2 + 4x - 1$
2. $f(x) = x^4 - 6x^2 + 2x + 3$
3. $f(x) = x + 1/x$
4. $f(x) = x + 3(1 - x)^{1/3}$
5. $f(x) = \sin x - \cos x$
6. $f(x) = \tan^{-1}(x^2)$
7. $f(x) = x^{4/3} + 4x^{1/3}$
8. $f(x) = xe^{-4x}$

In exercises 9–14, find all critical numbers and use the Second Derivative Test to determine all local extrema.

9. $f(x) = x^4 + 4x^3 - 1$
10. $f(x) = x^4 + 4x^2 + 1$
11. $f(x) = xe^{-x}$
12. $f(x) = e^{-x^2}$
13. $f(x) = \frac{x^2 - 5x + 4}{x}$
14. $f(x) = \frac{x^2 - 1}{x}$

In exercises 15–26, determine all significant features by hand and sketch a graph.

15. $f(x) = (x^2 + 1)^{2/3}$
16. $f(x) = x \ln x$
17. $f(x) = \frac{x^2}{x^2 - 9}$
18. $f(x) = \frac{x}{x + 2}$
19. $f(x) = \sin x + \cos x$
20. $f(x) = e^{-x} \sin x$
21. $f(x) = x^{3/4} - 4x^{1/4}$
22. $f(x) = x^{2/3} - 4x^{1/3}$
23. $f(x) = x|x|$
24. $f(x) = x^2|x|$
25. $f(x) = x^{1/5}(x + 1)$
26. $f(x) = \frac{\sqrt{x}}{1 + \sqrt{x}}$

In exercises 27–36, determine all significant features (approximately if necessary) and sketch a graph.

27. $f(x) = x^4 - 26x^3 + x$
28. $f(x) = 2x^4 - 11x^3 + 17x^2$
29. $f(x) = \sqrt[3]{2x^2 - 1}$
30. $f(x) = \sqrt{x^3 + 1}$
31. $f(x) = x^4 - 16x^3 + 42x^2 - 39.6x + 14$
32. $f(x) = x^4 + 32x^3 - 0.02x^2 - 0.8x$
33. $f(x) = x\sqrt{x^2 - 4}$
34. $f(x) = \frac{2x}{\sqrt{x^2 + 4}}$
35. $f(x) = \tan^{-1}\left(\frac{1}{x^2 - 1}\right)$
36. $f(x) = e^{-2x} \cos x$

In exercises 37–40, sketch a graph with the given properties.

37. $f(0) = 0$, $f'(x) > 0$ for $x < -1$ and $-1 < x < 1$, $f'(x) < 0$ for $x > 1$, $f''(x) > 0$ for $x < -1$, $0 < x < 1$ and $x > 1$, $f''(x) < 0$ for $-1 < x < 0$
38. $f(0) = 2$, $f'(x) > 0$ for all x , $f'(0) = 1$, $f''(x) > 0$ for $x < 0$, $f''(x) < 0$ for $x > 0$
39. $f(0) = 0$, $f(-1) = -1$, $f(1) = 1$, $f'(x) > 0$ for $x < -1$ and $0 < x < 1$, $f'(x) < 0$ for $-1 < x < 0$ and $x > 1$, $f''(x) < 0$ for $x < 0$ and $x > 0$
40. $f(1) = 0$, $f'(x) < 0$ for $x < 1$, $f'(x) > 0$ for $x > 1$, $f''(x) < 0$ for $x < 1$ and $x > 1$

41. Show that any cubic $f(x) = ax^3 + bx^2 + cx + d$ has one inflection point. Find conditions on the coefficients a – e that guarantee that the quartic $f(x) = ax^4 + bx^3 + cx^2 + dx + e$ has two inflection points.

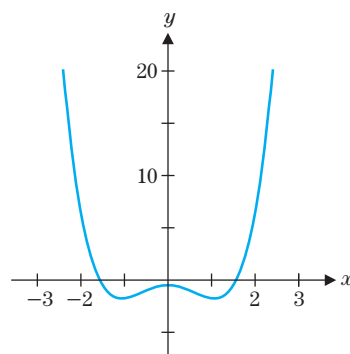
42. If f and g are functions with two derivatives for all x , $f(0) = g(0) = f'(0) = g'(0) = 0$, $f''(0) > 0$ and $g''(0) < 0$, state as completely as possible what can be said about whether $f(x) > g(x)$ or $f(x) < g(x)$.

43. Give an example of a function showing that the following statement is false. If the graph of $y = f(x)$ is concave down for all x , the equation $f(x) = 0$ has at least one solution.

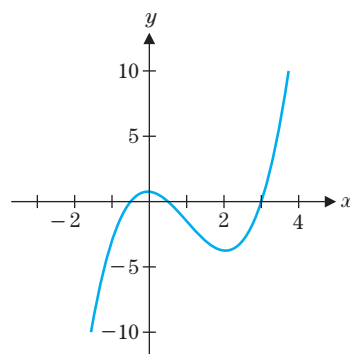
44. Determine whether the following statement is true or false. If $f(0) = 1$, $f''(x)$ exists for all x and the graph of $y = f(x)$ is concave down for all x , the equation $f(x) = 0$ has at least one solution.

In exercises 45 and 46, estimate the intervals of increase and decrease, the locations of local extrema, intervals of concavity and locations of inflection points.

45.



46.



47. Repeat exercises 45 and 46 if the given graph is of (a) f' or (b) f'' instead of f .
48. Prove Theorem 5.2 (the Second Derivative Test). (Hint: Think about what the definition of $f''(c)$ says when $f''(c) > 0$ or $f''(c) < 0$.)
49. Show that the function in example 5.4 can be written as $f(x) = (x^2 - 4)^2 - 6$. Conclude that the absolute minimum of f is -6 , occurring at $x = \pm 2$. Do a similar analysis with $g(x) = x^4 - 6x^2 + 1$.
50. For $f(x) = x^4 + bx^3 + cx^2 + dx + 2$, show that there are two inflection points if and only if $c < \frac{3}{8}b^2$. Show that the sum of the x -coordinates of the inflection points is $-b/2$.

APPLICATIONS

51. Suppose that $w(t)$ is the depth of water in a city's water reservoir at time t . Which would be better news at time $t = 0$, $w''(0) = 0.05$ or $w''(0) = -0.05$, or would you need to know the value of $w'(0)$ to determine which is better?
52. Suppose that $T(t)$ is a sick person's temperature at time t . Which would be better news at time t , $T''(0) = 2$ or $T''(0) = -2$, or would you need to know the value of $T'(0)$ and $T(0)$ to determine which is better?
53. Suppose that a company that spends AED x thousand on advertising sells AED $s(x)$ of merchandise, where $s(x) = -3x^3 + 270x^2 - 3600x + 18,000$. Find the value of x that maximize the rate of change of sales. (Hint: Read the question carefully!) Find the inflection point and explain why in advertising terms this is the "point of diminishing returns."
54. The number of units Q that a worker has produced in a day is related to the number of hours t since the work day began. Suppose that $Q(t) = -t^3 + 6t^2 + 12t$. Explain why $Q'(t)$ is a measure of the efficiency of the worker at time t . Find the time at which the worker's efficiency is a maximum. Explain why it is reasonable to call the inflection point the "point of diminishing returns."
55. Suppose that it costs a company $C(x) = 0.01x^2 + 40x + 3600$ dirhams to manufacture x units of a product. For this **cost function**, the **average cost function** is $\bar{C}(x) = \frac{C(x)}{x}$. Find the value of x that minimizes the average cost. The cost function can be related to the efficiency of the production process. Explain why a cost function that is concave down indicates better efficiency than a cost function that is concave up.
56. The antiarrhythmic medicine lidocaine slowly decays after entering the bloodstream. The plasma concentration t minutes after administering the drug can be modeled by
- $$c(t) = 92.8(-0.129e^{-t/6.55} + 0.218e^{-t/65.7} - 0.089e^{-t/13.3}).$$

Use a CAS to estimate the time of maximum concentration and inflection point ($t > 0$). Suppose that $f(t)$ represents the concentration of another type of medicine. If the graph of $f(t)$ has a similar shape and the same maximum point as the graph of $c(t)$ but the inflection point occurs at a larger value of t , would this medicine be more or less effective than lidocaine? Briefly explain.

57. A basic principle of physics is that light follows the path of minimum time. Assuming that the speed of light in the earth's atmosphere decreases as altitude decreases, argue that the path that light follows is concave down. Explain why this means that the setting Sun appears higher in the sky than it really is.



EXPLORATORY EXERCISES

- The linear approximation that we defined in section 4.1 is the line having the same location and the same slope as the function being approximated. Since two points determine a line, two requirements (point, slope) are all that a linear function can satisfy. However, a quadratic function can satisfy three requirements, since three points determine a parabola (and there are three constants in a general quadratic function $ax^2 + bx + c$). Suppose we want to define a **quadratic approximation** to $f(x)$ at $x = a$. Building on the linear approximation, the general form is $g(x) = f(a) + f'(a)(x - a) + c(x - a)^2$ for some constant c to be determined. In this way, show that $g(a) = f(a)$ and $g'(a) = f'(a)$. That is, $g(x)$ has the right position and slope at $x = a$. The third requirement is that $g(x)$ have the right concavity at $x = a$, so that $g''(a) = f''(a)$. Find the constant c that makes this true. Then, find such a quadratic approximation for each of the functions $\sin x$, $\cos x$ and e^x at $x = 0$. In each case, graph the original function, linear approximation and quadratic approximation, and describe how close the approximations are to the original functions.
- In this exercise, we explore a basic problem in genetics. Suppose that a species reproduces according to the following probabilities: p_0 is the probability of having no children, p_1 is the probability of having one offspring, p_2 is the probability of having two offspring, $\dots$, p_n is the probability of having n offspring and n is the largest number of offspring possible. Explain why for each i , we have $0 \leq p_i \leq 1$ and $p_0 + p_1 + p_2 + \dots + p_n = 1$. We define the function $F(x) = p_0 + p_1x + p_2x^2 + \dots + p_nx^n$. The smallest nonnegative solution of the equation $F(x) = x$ for $0 \leq x \leq 1$ represents the probability that the species becomes extinct. Show graphically that if $p_0 > 0$ and $F'(1) > 1$, then there is a solution of $F(x) = x$ with $0 < x < 1$. Thus, there is a positive probability of survival. However, if $p_0 > 0$ and $F'(1) < 1$, show that there are no solutions of $F(x) = x$ with $0 < x < 1$. (Hint: First show that F is increasing and concave up.)
- Give as complete a description of the graph of $f(x) = \frac{x+c}{x^2-1}$ as possible. In particular, find the values of c for which there are two critical points (or one critical point, or no critical points) and identify any extrema. Similarly, determine how the existence or not of inflection points depends on the value of c .

Overview of Curve Sketching

Graphing calculators and computer algebra systems are powerful aids in visualizing the graph of a function. However, they do not actually draw graphs. Instead, they plot points (albeit lots of them) and then connect the points as smoothly as possible. We have already seen that we must determine the appropriate window in which to draw a given graph, in order to see all of the significant features. We can accomplish this with some calculus.

We begin by summarizing the steps that you should take when trying to draw a graph of $y = f(x)$.

- **Domain:** Always determine the domain of f first.
- **Vertical Asymptotes:** For any isolated point not in the domain of f , check the limit of $f(x)$ as x approaches that point, to see if there is a vertical asymptote or a jump or removable discontinuity at that point.
- **First Derivative Information:** Determine where f is increasing and decreasing, and find any local extrema.
- **Vertical Tangent Lines:** At any isolated point not in the domain of f' , but in the domain of f , check the limit of $f'(x)$, to determine whether there is a vertical tangent line at that point.
- **Second Derivative Information:** Determine where the graph is concave up and concave down, and locate any inflection points.
- **Horizontal Asymptotes:** Check the limit of $f(x)$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.
- **Intercepts:** Locate x - and y -intercepts, if any. If this can't be done exactly, then do so approximately (e.g., using Newton's method).

We start with a very straightforward example.

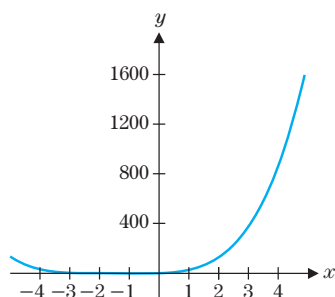


FIGURE 4.64a

$y = x^4 + 6x^3 + 12x^2 + 8x + 1$ (one view)

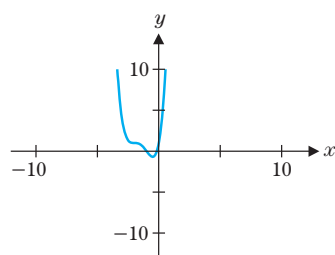
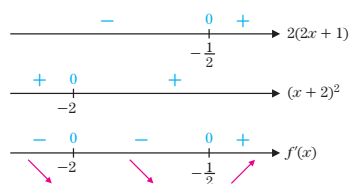


FIGURE 4.64b

$y = x^4 + 6x^3 + 12x^2 + 8x + 1$ (standard calculator view)



EXAMPLE 6.1 Drawing a Graph of a Polynomial

Draw a graph of $f(x) = x^4 + 6x^3 + 12x^2 + 8x + 1$, showing all significant features.

Solution One method commonly used by computer algebra systems and graphing calculators to determine the display window for a graph is to compute a set number of function values over a given standard range of x -values. The y -range is then chosen so that all of the calculated points can be displayed. This might result in a graph that looks like the one in Figure 4.64a. Another common method is to draw a graph in a fixed, default window. For instance, most graphing calculators use the default window defined by

$$-10 \leq x \leq 10 \quad \text{and} \quad -10 \leq y \leq 10.$$

Using this window, we get the graph shown in Figure 4.64b. Of course, these two graphs are very different and it's difficult to tell which, if either, of these is truly representative of the behavior of f . First, note that the domain of f is the entire real line. Further, since f is a polynomial, its graph doesn't have any vertical or horizontal asymptotes. Next, note that

$$f'(x) = 4x^3 + 18x^2 + 24x + 8 = 2(2x + 1)(x + 2)^2.$$

Drawing number lines for the individual factors of $f'(x)$, we have that

$$f'(x) \begin{cases} > 0, & \text{on } \left(-\frac{1}{2}, \infty\right) & f \text{ increasing.} \\ < 0, & \text{on } (-\infty, -2) \cup \left(-2, -\frac{1}{2}\right). & f \text{ decreasing.} \end{cases}$$

This also tells us that there is a local minimum at $x = -\frac{1}{2}$ and that there are no local maxima. Next, we have

$$f''(x) = 12x^2 + 36x + 24 = 12(x + 2)(x + 1).$$



Drawing number lines for the factors of $f'''(x)$, we have

$$f'''(x) \begin{cases} > 0, & \text{on } (-\infty, -2) \cup (-1, \infty) & \text{Concave up.} \\ < 0, & \text{on } (-2, -1). & \text{Concave down.} \end{cases}$$

From this, we see that there are inflection points at $x = -2$ and at $x = -1$. Finally, to find the x -intercepts, we need to solve $f(x) = 0$ approximately. Doing this (for instance, by using Newton's method or your calculator's solver), we find that there are two x -intercepts: $x = -1$ (exactly) and $x \approx -0.160713$. Notice that the significant x -values that we have identified are $x = -2$, $x = -1$ and $x = -\frac{1}{2}$. Computing the corresponding y -values from $y = f(x)$, we get the points $(-2, 1)$, $(-1, 0)$ and $(-\frac{1}{2}, -\frac{11}{16})$. We summarize the first and second derivative information in the number lines in the margin. In Figure 4.65, we include all of these important points by setting the x -range to be $-4 \leq x \leq 1$ and the y -range to be $-2 \leq y \leq 8$.

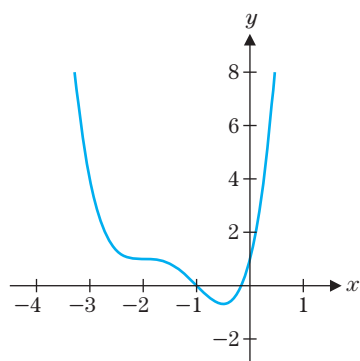


FIGURE 4.65

$$y = x^4 + 6x^3 + 12x^2 + 8x + 1$$

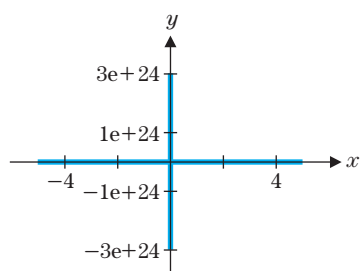


FIGURE 4.66a

$$y = \frac{x^2 - 3}{x^3}$$

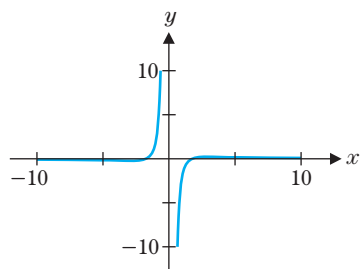


FIGURE 4.66b

$$y = \frac{x^2 - 3}{x^3}$$

EXAMPLE 6.2 Drawing a Graph of a Rational Function

Draw a graph of $f(x) = \frac{x^2 - 3}{x^3}$, showing all significant features.

Solution The default graph drawn by our computer algebra system appears in Figure 4.66a, while the graph drawn using the most common graphing calculator default window is seen in Figure 4.66b. This is arguably an improvement over Figure 4.66a, but this graph also leaves something to be desired, as we'll see.

First, observe that the domain of f includes all real numbers $x \neq 0$. Since $x = 0$ is an isolated point not in the domain of f , we consider

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x^2 - 3}{x^3} = -\infty \quad (6.1)$$

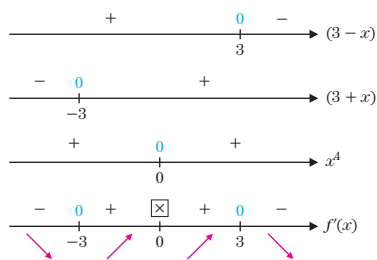
and

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x^2 - 3}{x^3} = \infty. \quad (6.2)$$

From (6.1) and (6.2), we see that the graph has a vertical asymptote at $x = 0$.

Next, we look for whatever information the first derivative will yield. We have

$$\begin{aligned} f'(x) &= \frac{2x(x^3) - (x^2 - 3)(3x^2)}{(x^3)^2} && \text{Quotient rule.} \\ &= \frac{x^2[2x^2 - 3(x^2 - 3)]}{x^6} && \text{Factor out } x^2. \\ &= \frac{9 - x^2}{x^4} && \text{Combine terms.} \\ &= \frac{(3 - x)(3 + x)}{x^4}. && \text{Factor difference of two squares.} \end{aligned}$$



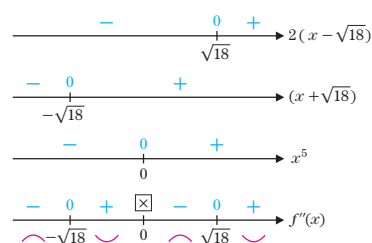
Looking at the individual factors in $f'(x)$, we have the number lines shown in the margin. Thus,

$$f'(x) \begin{cases} > 0, \text{ on } (-3, 0) \cup (0, 3) & f \text{ increasing.} \\ < 0, \text{ on } (-\infty, -3) \cup (3, \infty). & f \text{ decreasing.} \end{cases} \quad (6.3)$$

So, f has a local minimum at $x = -3$ and a local maximum at $x = 3$.

Next, we look at

$$\begin{aligned} f''(x) &= \frac{-2x(x^4) - (9 - x^2)(4x^3)}{(x^4)^2} && \text{Quotient rule.} \\ &= \frac{-2x^3[x^2 + (9 - x^2)(2)]}{x^8} && \text{Factor out } -2x^3. \\ &= \frac{-2(18 - x^2)}{x^5} && \text{Combine terms.} \\ &= \frac{2(x - \sqrt{18})(x + \sqrt{18})}{x^5}. && \text{Factor difference of two squares.} \end{aligned}$$



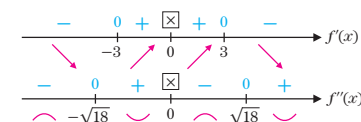
Looking at the individual factors in $f''(x)$, we obtain the number lines shown in the margin. Thus, we have

$$f''(x) \begin{cases} > 0, \text{ on } (-\sqrt{18}, 0) \cup (\sqrt{18}, \infty) & \text{Concave up.} \\ < 0, \text{ on } (-\infty, -\sqrt{18}) \cup (0, \sqrt{18}). & \text{Concave down.} \end{cases} \quad (6.4)$$

so that there are inflection points at $x = \pm\sqrt{18}$. (Why is there no inflection point at $x = 0$?)

To determine the limiting behavior as $x \rightarrow \pm\infty$, we consider

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{x^2 - 3}{x^3} \\ &= \lim_{x \rightarrow \infty} \left(\frac{1}{x} - \frac{3}{x^3} \right) = 0. \end{aligned} \quad (6.5)$$



Likewise, we have

$$\lim_{x \rightarrow -\infty} f(x) = 0. \quad (6.6)$$

So, the line $y = 0$ is a horizontal asymptote both as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. Finally, the x -intercepts are where

$$0 = f(x) = \frac{x^2 - 3}{x^3},$$

that is, at $x = \pm\sqrt{3}$. Notice that there are no y -intercepts, since $x = 0$ is not in the domain of the function. We now have all of the information that we need to draw a representative graph. With some experimentation, you can set the x - and y -ranges so that most of the significant features of the graph (i.e., vertical and horizontal asymptotes, local extrema, inflection points, etc.) are displayed, as in Figure 4.67, which is consistent with all of the information that we accumulated about the function in (6.1)–(6.6). Although the existence of the inflection points is clearly indicated by the change in concavity, their precise location is as yet a bit fuzzy in this graph. However, both vertical and horizontal asymptotes and the local extrema are clearly indicated, something that cannot be said about either of Figures 4.66a or 4.66b. ■

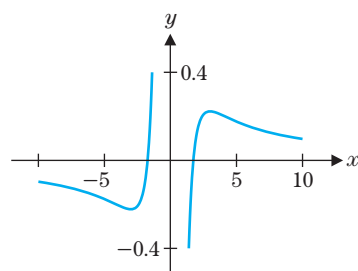


FIGURE 4.67

$$y = \frac{x^2 - 3}{x^3}$$

In example 6.3, there are multiple vertical asymptotes, only one extremum and no inflection points.

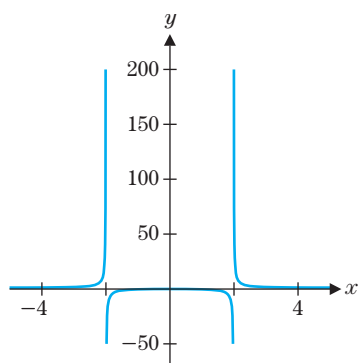


FIGURE 4.68a

$$y = \frac{x^2}{x^2 - 4}$$

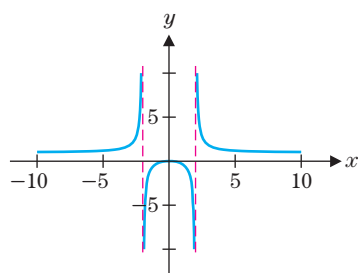


FIGURE 4.68b

$$y = \frac{x^2}{x^2 - 4}$$

EXAMPLE 6.3 A Graph with Two Vertical Asymptotes

Draw a graph of $f(x) = \frac{x^2}{x^2 - 4}$ showing all significant features.

Solution The default graph produced by our computer algebra system is seen in Figure 4.68a, while the default graph drawn by most graphing calculators looks like the graph seen in Figure 4.68b. Notice that the domain of f includes all x except $x = \pm 2$ (since the denominator is zero at $x = \pm 2$). Figure 4.68b suggests that there are vertical asymptotes at $x = \pm 2$, but let's establish this carefully. We have

$$\lim_{x \rightarrow 2^+} \frac{x^2}{x^2 - 4} = \lim_{x \rightarrow 2^+} \frac{x^2}{\underset{+}{(x-2)}\underset{+}{(x+2)}} = \infty. \quad (6.7)$$

Similarly, we get

$$\lim_{x \rightarrow 2^-} \frac{x^2}{x^2 - 4} = -\infty, \quad \lim_{x \rightarrow -2^+} \frac{x^2}{x^2 - 4} = -\infty \quad (6.8)$$

and

$$\lim_{x \rightarrow -2^-} \frac{x^2}{x^2 - 4} = \infty. \quad (6.9)$$

Thus, there are vertical asymptotes at $x = \pm 2$. Next, we have

$$f'(x) = \frac{2x(x^2 - 4) - x^2(2x)}{(x^2 - 4)^2} = \frac{-8x}{(x^2 - 4)^2}.$$

Since the denominator is positive for $x \neq \pm 2$, it is a simple matter to see that

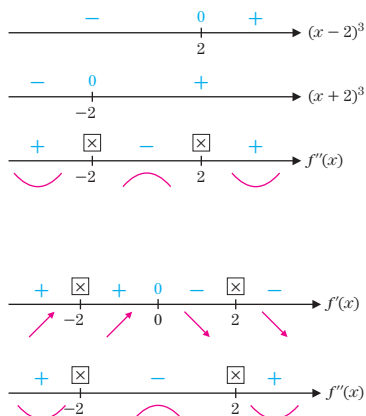
$$f'(x) \begin{cases} > 0, & \text{on } (-\infty, -2) \text{ and } (-2, 0) & f \text{ increasing.} \\ < 0, & \text{on } (0, 2) \text{ and } (2, \infty). & f \text{ decreasing.} \end{cases} \quad (6.10)$$

In particular, notice that the only critical number is $x = 0$ (since $x = -2, 2$ are not in the domain of f). Thus, the only local extremum is the local maximum located at $x = 0$. Next, we have

$$\begin{aligned} f''(x) &= \frac{-8(x^2 - 4)^2 + (8x)2(x^2 - 4)^1(2x)}{(x^2 - 4)^4} && \text{Quotient rule.} \\ &= \frac{8(x^2 - 4)[-(x^2 - 4) + 4x^2]}{(x^2 - 4)^4} && \text{Factor out } 8(x^2 - 4). \\ &= \frac{8(3x^2 + 4)}{(x^2 - 4)^3} && \text{Combine terms.} \\ &= \frac{8(3x^2 + 4)}{(x - 2)^3(x + 2)^3}. && \text{Factor difference of two squares.} \end{aligned}$$

Since the numerator is positive for all x , we need only consider the factors in the denominator, as seen in the margin. We then have

$$f''(x) \begin{cases} > 0, & \text{on } (-\infty, -2) \cup (2, \infty) & \text{Concave up.} \\ < 0, & \text{on } (-2, 2). & \text{Concave down.} \end{cases} \quad (6.11)$$



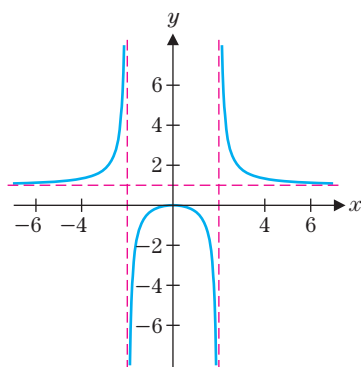


FIGURE 4.69

$$y = \frac{x^2}{x^2 - 4}$$

However, since $x = 2, -2$ are not in the domain of f , there are no inflection points. It is an easy exercise to verify that

$$\lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 4} = 1 \quad (6.12)$$

and

$$\lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - 4} = 1. \quad (6.13)$$

From (6.12) and (6.13), we have that $y = 1$ is a horizontal asymptote, both as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. Finally, we observe that the only x -intercept is at $x = 0$. We summarize the information in (6.7)–(6.13) in the graph seen in Figure 4.69. ■

In example 6.4, we need to use computer-generated graphs, as well as a rootfinding method to determine the behavior of the function.

EXAMPLE 6.4 Graphing Where the Domain and Extrema Must Be Approximated

Draw a graph of $f(x) = \frac{1}{x^3 + 3x^2 + 3x + 3}$ showing all significant features.

Solution The default graph drawn by most graphing calculators and computer algebra systems looks something like the one shown in Figure 4.70. We use some calculus to refine this.

Since f is a rational function, it is defined for all x , except for where the denominator is zero, that is, where

$$g(x) = x^3 + 3x^2 + 3x + 3 = 0.$$

From the graph of $y = g(x)$ in Figure 4.71, we see that g has only one zero, around $x = -2$. We can verify that this is the only zero, since

$$g'(x) = \frac{d}{dx}(x^3 + 3x^2 + 3x + 3) = 3x^2 + 6x + 3 = 3(x + 1)^2 \geq 0.$$

You can get the approximate zero $x = a \approx -2.25992$ using Newton's method or your calculator's solver. We can use the graph in Figure 4.71 to help us compute the limits

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} \frac{1}{x^3 + 3x^2 + 3x + 3} = \infty \quad (6.14)$$

and

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^-} \frac{1}{x^3 + 3x^2 + 3x + 3} = -\infty. \quad (6.15)$$

From (6.14) and (6.15), f has a vertical asymptote at $x = a$. Turning to the derivative

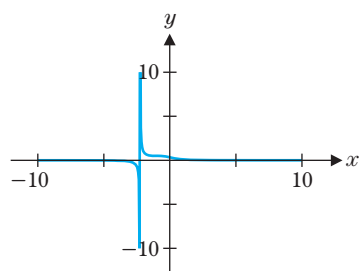


FIGURE 4.70

$$y = \frac{1}{x^3 + 3x^2 + 3x + 3}$$

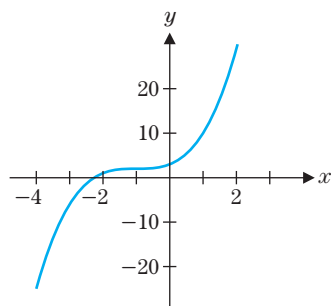


FIGURE 4.71

$$y = x^3 + 3x^2 + 3x + 3$$

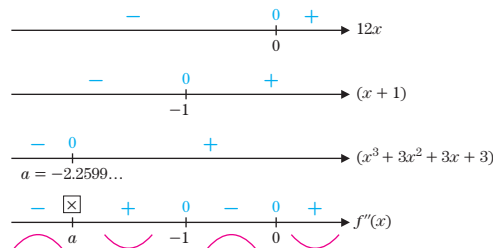
information, we have

$$\begin{aligned}
 f'(x) &= -(x^3 + 3x^2 + 3x + 3)^{-2}(3x^2 + 6x + 3) \\
 &= -3 \left[\frac{(x+1)^2}{(x^3 + 3x^2 + 3x + 3)^2} \right] \\
 &= -3 \left(\frac{x+1}{x^3 + 3x^2 + 3x + 3} \right)^2 \\
 &< 0, \text{ for } x \neq a \text{ or } -1
 \end{aligned} \tag{6.16}$$

and $f'(-1) = 0$. Thus, f is decreasing for $x < a$ and $x > a$. Also, notice that the only critical number is $x = -1$, but since f is decreasing everywhere except at $x = a$, there are no local extrema. Turning to the second derivative, we get

$$\begin{aligned}
 f''(x) &= -6 \left(\frac{x+1}{x^3 + 3x^2 + 3x + 3} \right) \frac{1(x^3 + 3x^2 + 3x + 3) - (x+1)(3x^2 + 6x + 3)}{(x^3 + 3x^2 + 3x + 3)^2} \\
 &= \frac{-6(x+1)}{(x^3 + 3x^2 + 3x + 3)^3} (-2x^3 - 6x^2 - 6x) \\
 &= \frac{12x(x+1)(x^2 + 3x + 3)}{(x^3 + 3x^2 + 3x + 3)^3}.
 \end{aligned}$$

Since $(x^2 + 3x + 3) > 0$ for all x (why is that?), we need not consider this factor. Considering the remaining factors, we have the number lines shown here.



Thus, we have that

$$f''(x) \begin{cases} > 0, \text{ on } (a, -1) \text{ and } (0, \infty) & \text{Concave up.} \\ < 0, \text{ on } (-\infty, a) \text{ and } (-1, 0). & \text{Concave down.} \end{cases} \tag{6.17}$$

It now follows that there are inflection points at $x = 0$ and at $x = -1$. Notice that in Figure 4.70, the concavity information is not very clear and the inflection points are difficult to discern.

We note the obvious fact that the function is never zero and hence, there are no x -intercepts. Finally, we consider the limits

$$\lim_{x \rightarrow \infty} \frac{1}{x^3 + 3x^2 + 3x + 3} = 0 \tag{6.18}$$

and

$$\lim_{x \rightarrow -\infty} \frac{1}{x^3 + 3x^2 + 3x + 3} = 0. \tag{6.19}$$

Using all of the information in (6.14)–(6.19), we draw the graph seen in Figure 4.72. Here, we can clearly see the vertical and horizontal asymptotes, the inflection points and the fact that the function is decreasing across its entire domain. ■

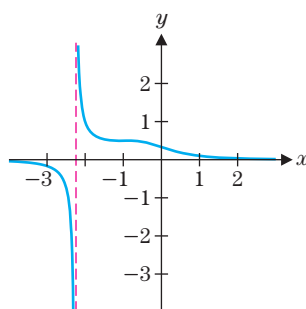


FIGURE 4.72
 $y = \frac{1}{x^3 + 3x^2 + 3x + 3}$

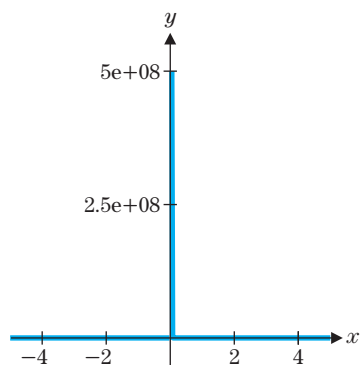


FIGURE 4.73a

$$y = e^{1/x}$$

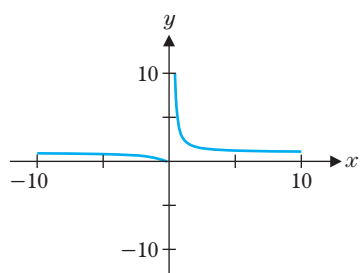


FIGURE 4.73b

$$y = e^{1/x}$$

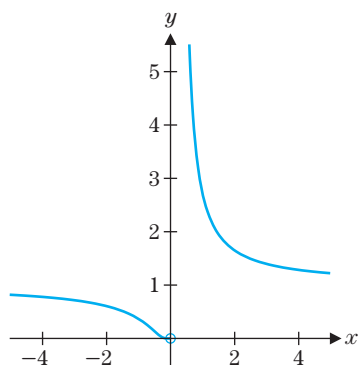


FIGURE 4.74

$$y = e^{1/x}$$

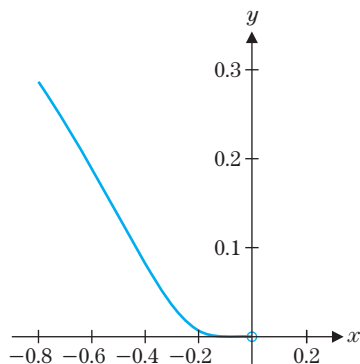


FIGURE 4.75

$$y = e^{1/x}$$

EXAMPLE 6.5 Graphing Where Some Features Are Difficult to See

Draw a graph of $f(x) = e^{1/x}$ showing all significant features.

Solution The default graph produced by our computer algebra system is not particularly helpful. (See Figure 4.73a.) The default graph produced by most graphing calculators (see Figure 4.73b) is certainly better, but we can't be sure this is adequate without further analysis. First, notice that the domain of f is $(-\infty, 0) \cup (0, \infty)$. Thus, we consider

$$\lim_{x \rightarrow 0^+} e^{1/x} = \infty, \quad (6.20)$$

since $1/x \rightarrow \infty$ as $x \rightarrow 0^+$. Also, since $1/x \rightarrow -\infty$ as $x \rightarrow 0^-$ (and $e^t \rightarrow 0$ as $t \rightarrow -\infty$), we have

$$\lim_{x \rightarrow 0^-} e^{1/x} = 0. \quad (6.21)$$

From (6.20) and (6.21), there is a vertical asymptote at $x = 0$, but an unusual one, in that $f(x) \rightarrow \infty$ on one side of 0 and $f(x) \rightarrow 0$ on the other side. Next,

$$\begin{aligned} f'(x) &= e^{1/x} \frac{d}{dx} \left(\frac{1}{x} \right) \\ &= e^{1/x} \left(\frac{-1}{x^2} \right) < 0, \text{ for all } x \neq 0, \end{aligned}$$

since $e^{1/x} > 0$, for all $x \neq 0$. Hence, f is decreasing for all $x \neq 0$. We also have

$$\begin{aligned} f''(x) &= e^{1/x} \left(\frac{-1}{x^2} \right) \left(\frac{-1}{x^2} \right) + e^{1/x} \left(\frac{2}{x^3} \right) \\ &= e^{1/x} \left(\frac{1}{x^4} + \frac{2}{x^3} \right) = e^{1/x} \left(\frac{1 + 2x}{x^4} \right) \\ &\begin{cases} < 0, & \text{on } \left(-\infty, -\frac{1}{2} \right) & \text{Concave down.} \\ > 0, & \text{on } \left(-\frac{1}{2}, 0 \right) \cup (0, \infty). & \text{Concave up.} \end{cases} \end{aligned}$$

Since $x = 0$ is not in the domain of f , the only inflection point is at $x = -\frac{1}{2}$. Next, note that

$$\lim_{x \rightarrow \infty} e^{1/x} = 1,$$

since $1/x \rightarrow 0$ as $x \rightarrow \infty$ and $e^t \rightarrow 1$ as $t \rightarrow 0$. Likewise,

$$\lim_{x \rightarrow -\infty} e^{1/x} = 1.$$

So, $y = 1$ is a horizontal asymptote, both as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. Finally, since

$$e^{1/x} > 0,$$

for all $x \neq 0$, there are no x -intercepts. Note that in almost any graph you draw, it is difficult to see all of the features of the function, since the inflection point $\left(-\frac{1}{2}, e^{-2} \right)$ is so close to the x -axis. Since the horizontal asymptote is the line $y = 1$, it is difficult to see both of these features on the same graph (without drawing the graph on a very large piece of paper). We settle for the graph seen in Figure 4.74, which shows all of the features except the inflection point and the concavity on the interval $\left(-\frac{1}{2}, 0 \right)$. To clearly see the behavior near the inflection point, we draw a graph that is zoomed-in on the area of the inflection point. (See Figure 4.75.) Here, while we have resolved the problem of the concavity near $x = 0$ and the inflection point, we have lost the details of the "big picture."

In our final example, we consider the graph of a function that is the sum of a trigonometric function and a polynomial.

EXAMPLE 6.6 Graphing the Sum of a Polynomial and a Trigonometric Function

Draw a graph of $f(x) = \cos x - x$, showing all significant features.

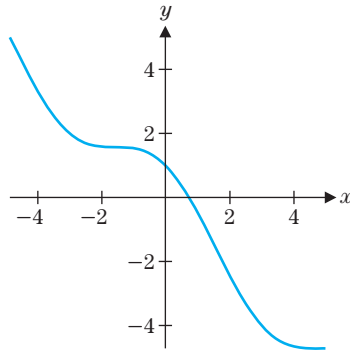


FIGURE 4.76a
 $y = \cos x - x$

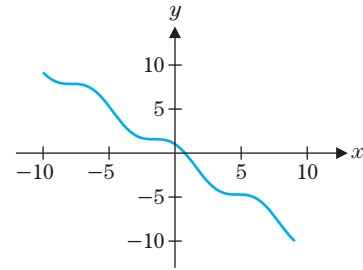


FIGURE 4.76b
 $y = \cos x - x$

Solution The default graph provided by our computer algebra system can be seen in Figure 4.76a. The graph produced by most graphing calculators looks like that in Figure 4.76b. Since the domain of f is the entire real line, there are no vertical asymptotes. Next, we have

$$f'(x) = -\sin x - 1 \leq 0, \quad \text{for all } x. \quad (6.22)$$

Further, $f'(x) = 0$ if and only if $\sin x = -1$. So, there are critical numbers (here, these are all locations of horizontal tangent lines), but since $f'(x)$ does not change sign, there are *no* local extrema. Even so, it is still of interest to find the locations of the horizontal tangent lines. Recall that

$$\sin x = -1 \quad \text{for } x = \frac{3\pi}{2}$$

and more generally, for
$$x = \frac{3\pi}{2} + 2n\pi,$$

for any integer n . Next, we see that

$$f''(x) = -\cos x$$

and on the interval $[0, 2\pi]$, we have

$$\cos x \begin{cases} > 0, & \text{on } \left[0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right] \\ < 0, & \text{on } \left(\frac{\pi}{2}, \frac{3\pi}{2}\right). \end{cases}$$

$$\text{So, } f''(x) = -\cos x \begin{cases} < 0, & \text{on } \left[0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right] & \text{Concave down.} \\ > 0, & \text{on } \left(\frac{\pi}{2}, \frac{3\pi}{2}\right). & \text{Concave up.} \end{cases} \quad (6.23)$$

Outside of $[0, 2\pi]$, $f''(x)$ simply repeats this pattern. In particular, this says that the graph has infinitely many inflection points, located at odd multiples of $\pi/2$.

To determine the behavior as $x \rightarrow \pm\infty$, we examine the limits

$$\lim_{x \rightarrow \infty} (\cos x - x) = -\infty \quad (6.24)$$

$$\text{and } \lim_{x \rightarrow -\infty} (\cos x - x) = \infty, \quad (6.25)$$

since $-1 \leq \cos x \leq 1$, for all x and since $\lim_{x \rightarrow \infty} x = \infty$.

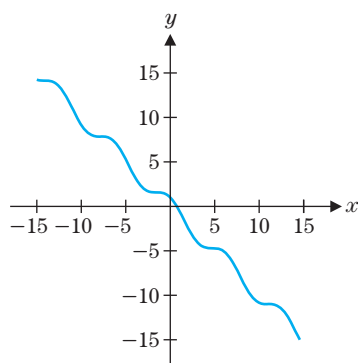


FIGURE 4.77
 $y = \cos x - x$

Finally, to determine the x -intercept(s), we need to solve

$$f(x) = \cos x - x = 0.$$

This can't be solved exactly, however. Since $f'(x) \leq 0$ for all x and Figures 4.76a and 4.76b show a zero around $x = 1$, there is only one zero and we must approximate this. (Use Newton's method or your calculator's solver.) We get $x \approx 0.739085$ as an approximation to the only x -intercept. Assembling all of the information in (4.22)–(4.25), we can draw the graph seen in Figure 4.77. Notice that Figure 4.76b shows the behavior just as clearly as Figure 4.77, but for a smaller range of x - and y -values. Which of these is more “representative” is open to discussion. ■

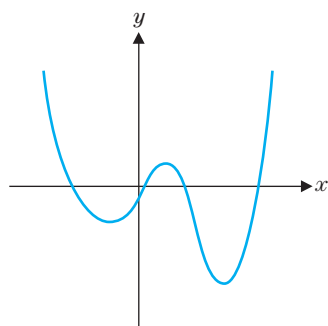
BEYOND FORMULAS

The main characteristic of the examples in sections 4.4–4.6 is the interplay between graphing and equation solving. To analyze the graph of a function, you will go back and forth several times between solving equations (for critical numbers and inflection points and so on) and identifying graphical features of interest. The equation solving may lead you to uncover hidden features of the graph.

EXERCISES 4.6

WRITING EXERCISES

- We have talked about sketching representative graphs, but it is often impossible to draw a graph correctly to scale that shows all of the properties we might be interested in. For example, try to generate a computer or calculator graph that shows all three local extrema of $x^4 - 25x^3 - 2x^2 + 80x - 3$. When two extrema have y -coordinates of approximately -60 and 50 , it takes a very large graph to also show a point with $y = -40,000$! If an accurate graph cannot show all the points of interest, perhaps a freehand sketch like the one shown below is needed.



There is no scale shown on the graph because we have distorted different portions of the graph in an attempt to show all of the interesting points. Discuss the relative merits of an “honest” graph with a consistent scale but not showing all the points of interest versus a caricature graph that distorts the scale but does show all the points of interest.

- Explain how the graph of $f(x) = \cos x - x$ in example 6.6 relates to the graphs of $y = \cos x$ and $y = -x$. Based on this discussion, explain how to sketch the graph of $y = x + \sin x$.

In exercises 1–22, graph the function and completely discuss the graph as in example 6.2.

- | | |
|--|-------------------------------------|
| 1. $f(x) = x^3 - 3x^2 + 3x$ | 2. $f(x) = x^4 - 3x^2 + 2$ |
| 3. $f(x) = x^5 - 2x^3 + 1$ | 4. $f(x) = x^4 + 4x^3 - 1$ |
| 5. $f(x) = x + \frac{4}{x}$ | 6. $f(x) = \frac{x^2 - 1}{x}$ |
| 7. $f(x) = \frac{x^2 + 4}{x^3}$ | 8. $f(x) = \frac{x - 4}{x^3}$ |
| 9. $f(x) = \frac{2x}{x^2 - 1}$ | 10. $f(x) = \frac{3x^2}{x^2 + 1}$ |
| 11. $f(x) = x + \sin x$ | 12. $f(x) = \sin x - \cos x$ |
| 13. $f(x) = x \ln x$ | 14. $f(x) = x \ln x^2$ |
| 15. $f(x) = \sqrt{x^2 + 1}$ | 16. $f(x) = \sqrt{2x - 1}$ |
| 17. $f(x) = \sqrt[3]{x^3 - 3x^2 + 2x}$ | 18. $f(x) = \sqrt{x^3 - 3x^2 + 2x}$ |
| 19. $f(x) = x^{5/3} - 5x^{2/3}$ | 20. $f(x) = x^3 - \frac{3}{400}x$ |
| 21. $f(x) = e^{-2/x}$ | 22. $f(x) = e^{1/x^2}$ |

In exercises 23–36, determine all significant features (approximately if necessary) and sketch a graph.

- $f(x) = \frac{1}{x^3 - 3x^2 - 9x + 1}$
- $f(x) = \frac{1}{x^3 + 3x^2 + 4x + 1}$
- $f(x) = (x^3 - 3x^2 + 2x)^{2/3}$
- $f(x) = x^6 - 10x^5 - 7x^4 + 80x^3 + 12x^2 - 192x$
- $f(x) = \frac{x^2 + 1}{3x^2 - 1}$
- $f(x) = \frac{5x}{x^3 - x + 1}$

29. $f(x) = x^2 \sqrt{x^2 - 9}$ 30. $f(x) = \sqrt[3]{2x^2 - 1}$
 31. $f(x) = e^{-2x} \sin x$ 32. $f(x) = \sin x - \frac{1}{2} \sin 2x$
 33. $f(x) = x^4 - 16x^3 + 42x^2 - 39.6x + 14$
 34. $f(x) = x^4 + 32x^3 - 0.02x^2 - 0.8x$
 35. $f(x) = \frac{25 - 50\sqrt{x^2 + 0.25}}{x}$ 36. $f(x) = \tan^{-1}\left(\frac{1}{x^2 - 1}\right)$

In exercises 37–42, the “family of functions” contains a parameter c . The value of c affects the properties of the functions. Determine what differences, if any, there are for c being zero, positive or negative. Then determine what the graph would look like for very large positive c 's and for very large negative c 's.

37. $f(x) = x^4 + cx^2$ 38. $f(x) = x^4 + cx^2 + x$
 39. $f(x) = \frac{x^2}{x^2 + c^2}$ 40. $f(x) = e^{-x^2/c}$
 41. $f(x) = \sin(cx)$ 42. $f(x) = x^2 \sqrt{c^2 - x^2}$

A function f has a slant asymptote $y = mx + b$ ($m \neq 0$) if $\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0$ and/or $\lim_{x \rightarrow -\infty} [f(x) - (mx + b)] = 0$. In exercises 43–48, find the slant asymptote. (Use long division to rewrite the function.) Then, graph the function and its asymptote on the same axes.

43. $f(x) = \frac{3x^2 - 1}{x}$ 44. $f(x) = \frac{3x^2 - 1}{x - 1}$
 45. $f(x) = \frac{x^3 - 2x^2 + 1}{x^2}$ 46. $f(x) = \frac{x^3 - 1}{x^2 - 1}$
 47. $f(x) = \frac{x^4}{x^3 + 1}$ 48. $f(x) = \frac{x^4 - 1}{x^3 + x}$

In exercises 49–52, find a function whose graph has the given asymptotes.

49. $x = 1$, $x = 2$ and $y = 3$ 50. $x = -1$, $x = 1$ and $y = 0$
 51. $x = -1$, $x = 1$, $y = -2$ and $y = 2$
 52. $x = 1$, $y = 2$ and $x = 3$

53. It can be useful to identify asymptotes other than vertical and horizontal. For example, the parabola $y = x^2$ is an asymptote of $f(x)$ if $\lim_{x \rightarrow \infty} [f(x) - x^2] = 0$ and/or $\lim_{x \rightarrow -\infty} [f(x) - x^2] = 0$. Show that x^2 is an asymptote of $f(x) = \frac{x^4 - x^2 + 1}{x^2 - 1}$. Graph $y = f(x)$ and zoom out until the graph looks like a parabola. (Note: The effect of zooming out is to emphasize large values of x .)

54. For each function, find a polynomial $p(x)$ such that $\lim_{x \rightarrow \infty} [f(x) - p(x)] = 0$.

- (a) $\frac{x^4}{x + 1}$ (b) $\frac{x^5 - 1}{x + 1}$ (c) $\frac{x^6 - 2}{x + 1}$

Show by zooming out that $f(x)$ and $p(x)$ look similar for large x .

55. Find all extrema and inflection points, and sketch the graphs of $y = \sinh x = \frac{e^x - e^{-x}}{2}$ and $y = \cosh x = \frac{e^x + e^{-x}}{2}$.
 56. On the same axes as the graphs of exercise 55, sketch in the graphs of $y = \frac{1}{2}e^x$ and $y = \frac{1}{2}e^{-x}$. Explain why these graphs serve as an envelope for the graphs in exercise 55. (Hint: As $x \rightarrow \pm\infty$, what happens to e^x and e^{-x} ?)

APPLICATIONS

57. In a variety of applications, researchers model a phenomenon whose graph starts at the origin, rises to a single maximum and then drops off to a horizontal asymptote of $y = 0$. For example, the probability density function of events such as the time from conception to birth of an animal and the amount of time surviving after contracting a fatal disease might have these properties. Show that the family of functions xe^{-bx} has these properties for all positive constants b . What effect does b have on the location of the maximum? In the case of the time since conception, what would b represent? In the case of survival time, what would b represent?
58. The “FM” in FM radio stands for **frequency modulation**, a method of transmitting information encoded in a radio wave by modulating (or varying) the frequency. A basic example of such a modulated wave is $f(x) = \cos(10x + 2 \cos x)$. Use computer-generated graphs of $f(x)$, $f'(x)$ and $f''(x)$ to try to locate all local extrema of $f(x)$.
59. The angle for a field goal kicked from a hash mark at a distance of x meters is $A = \tan^{-1}\left(\frac{29.25}{x}\right) - \tan^{-1}\left(\frac{10.75}{x}\right)$. Find x to maximize the angle A . A 4.5-meter penalty increases x from 18.3 to 22.9. How does this change A ?
60. A knuckleball thrown with rotation rate ω (in rad/s) has lateral position $x(t) = \frac{2.5}{\omega}t - \frac{2.5}{4\omega^2} \sin 4\omega t$ at time t , for $0 \leq t \leq 0.68$. Explore the effect on the graph of changing $\omega \geq 0$.

EXPLORATORY EXERCISES

1. One of the natural enemies of the balsam fir tree is the spruce budworm, which attacks the leaves of the fir tree in devastating outbreaks. Define $N(t)$ to be the number of worms on a particular tree at time t . A mathematical model of the population dynamics of the worm must include a term to indicate the worm's death rate due to its predators (e.g., birds). The form of this term is often taken to be $\frac{B[N(t)]^2}{A^2 + [N(t)]^2}$ for positive constants A and B . Graph the functions $\frac{x^2}{4 + x^2}$, $\frac{2x^2}{1 + x^2}$, $\frac{x^2}{9 + x^2}$ and $\frac{3x^2}{1 + x^2}$ for $x > 0$. Based on these graphs, discuss why $\frac{B[N(t)]^2}{A^2 + [N(t)]^2}$ is a plausible model for the death rate by predation. What role do the constants A and B play? The possible stable population levels for the spruce budworms are

determined by intersections of the graphs of $y = r(1 - x/k)$ and $y = \frac{x}{1+x^2}$. Here, $x = N/A$, r is proportional to the birthrate of the budworms and k is determined by the amount of food available to the budworms. Note that $y = r(1 - x/k)$ is a line with y -intercept r and x -intercept k . How many solutions are there to the equation $r(1 - x/k) = \frac{x}{1+x^2}$? (Hint: The answer depends on the values of r and k .) One current theory is that outbreaks are caused in situations where there are three solutions and the population of budworms jumps from a small population to a large population.

- Suppose that f is a function with two derivatives and that $f(a) = f'(a) = 0$ but $f''(a) \neq 0$ for some number a . Show that $f(x)$ has a local extremum at $x = a$. Next, suppose that f is a function with three derivatives and that $f(a) = f'(a) = f''(a) = 0$ but $f'''(a) \neq 0$ for some number a . Show that $f(x)$ does *not* have a local extremum at $x = a$. Generalize your work to the case where $f^{(k)}(a) = 0$ for $k = 0, 1, \dots, n-1$, but $f^{(n)}(a) \neq 0$, keeping in mind that there are different conclusions depending on whether n is odd or even. Use this result to determine whether $f(x) = x \sin x^2$ or $g(x) = x^2 \sin(x^2)$ has a local extremum at $x = 0$.